

Definition 0.11. For any $T \in \mathcal{L}(V, W)$, set $\text{null } T = \{ v \mid Tv = 0 \}$ is called the **null space** of T .

This is also called the **kernal** of T in algebra.

Theorem 0.13. For any $T \in \mathcal{L}(V, W)$, $\text{null } T$ is a subspace of V .

Proof.

- We have $0 \in \text{null } T$ since $T0 = 0$, which is the property of linear transformation.
- For any $a, b \in \text{null } T$, we have $0 = Ta + Tb = T(a + b)$, so $a + b \in \text{null } T$.
- For any $Ta \in \text{null } T$ and $\lambda \in F$, we have $\lambda Ta = T(\lambda a)$, so $\lambda a \in \text{null } T$.

□

Definition 0.15. For any $T \in \mathcal{L}(V, W)$, set $\text{range } T = T(V) = \{ Tv \mid v \in V \}$ is called the **range** of T .

This is also called the **image** of T in math.

Theorem 0.18. For any $T \in \mathcal{L}(V, W)$, $\text{range } T$ is a subspace of W .

Proof.

- We have $T(0) = 0 \in \text{range } T$.
- For any $Ta, Tb \in \text{range } T$, $Ta + Tb = T(a + b) \in \text{range } T$.
- For any $Ta \in \text{range } T$ and $\lambda \in F$, $\lambda Ta = T(\lambda a) \in \text{range } T$.

□

Theorem 0.21. Suppose V is finite and $T \in \mathcal{L}(V, W)$, then $\text{range } T$ is finite, and

$$\dim V = \dim \text{null } T + \dim \text{range } T$$

Proof. Consider the basis $v_0, \dots, v_k$ of $\text{null } T$, and the basis $v_{k+1}, \dots, v_n$ of V that expand from $v_0, \dots, v_k$. We will show that $T(v_{k+1}), \dots, T(v_n)$ is the basis of $\text{range } T$.

We first show that $T(v_{k+1}), \dots, T(v_n)$ is linear independent. If it is linear independent, then

$$\begin{aligned}
& \lambda_1 T(v_{k+1}) + \cdots + \lambda_i T(v_{k+i}) \\
&= T(\lambda_1 v_{k+1} + \cdots + \lambda_i v_{k+i}) \\
&= 0
\end{aligned}$$

That means a linear combination of v_{k+i} is in $\text{null } T$, which is $\text{span}(v_0, \dots, v_k)$, therefore the basis $v_0, \dots, v_n$ is linear dependent.

Then we show that $T(v_{k+1}), \dots, T(v_n)$ spans $\text{range } T$. For any $Tv \in \text{range } T$, there must be $v \in V$ such that $Tv = Tv$, then v can be written in form of the linear combination of $v_0, \dots, v_n$, and then $Tv = T(\lambda_0 v_0 + \cdots + \lambda_n v_n)$. We can drop all terms with v_i where $i \leq k$, since they are in $\text{null } T$, so Tv is now represented by a linear combination of $T(v_{k+i})$ for all $0 < i \leq n - k$, therefore, it is a basis of $\text{range } T$ and $\dim \text{range } T$ is finite.

Finally, $\dim V = \dim \text{null } T + \dim \text{range } T$. □

Definition 3.16 (Notation: $v + U$). Let $v \in V$ and $U \subseteq V$, then $v + U = \{ v + u \mid u \in U \}$.

Such sets also called *coset* in group theory.

Definition 3.97 (Translate). Let $v \in V$ and $U \subseteq V$, we say $v + U$ is a translate of U .

Definition 3.98 (Quotient Space). Let $U \subseteq V$ a subspace, then the quotient space V/U is a set with translates of U , that is:

$$V/U = \{ v + U \mid v \in V \}$$

Theorem 3.101. Let $U \subseteq V$ a subspace and $v, w \in V$, then the following statements are equivalent.

1. $v - w \in U$
2. $v + U = w + U$
3. $(v + U) \cap (w + U) \neq \emptyset$

Proof.

- If $v - w \in U$, for any $v + u \in v + U$, we have $v + u = v + (v - w) - (v - w) + u = v - w + w + u = w + (v - w) + u \in w + U$ since $v - w \in U$. Similarly, for any $w + u \in w + U$, we have $w + u = w + (v - w) - (v - w) + u = v - v + w + u = v - (v - w) + u = v + (-(v - w) + u) \in v + U$.

- If $v+U = w+U$, then $v = w+u$ since $v \in v+U$, therefore $v-w = u \in U$.
- if $v + U = w + U$, then $(v + U) \cap (w + U) = v + U = w + U \neq \emptyset$
- If $(v+U) \cap (w+U) \neq \emptyset$, then for any $v+u_0 = w+u_1 \in (v+U) \cap (w+U)$, we have $(v - w) + (u_0 - u_1) = 0$ and then $v - w = u_1 - u_0 \in U$, so $v + U = w + U$.

□

Definition 3.102. Let $U \subseteq V$, then addition and scalar multiplication on V/U is defined by:

$$(v + U) + (w + U) = (v + w) + U$$

$$\lambda(v + U) = (\lambda v) + U$$

Theorem 3.103. Let $U \subseteq V$ a subspace, then V/U is a vector space with addition and scalar multiplication we defined in previous definition.

Proof. We must first show that the addition and the scalar multiplication we introduce are functions.

For any $a, b, c, d \in V$, we will show $(a+b)+U = (c+d)+U$ if $a+U = c+U$ and $b+U = d+U$. We can show $(a+b) - (c+d) \in U$ by $a - c \in U$ and $b - d \in U$.

For any $v, w \in V$ and $\lambda \in F$, we will show $(\lambda v) + U = (\lambda w) + U$ if $v+U = w+U$. We know $v - w \in U$, then $\lambda(v - w) = \lambda v - \lambda w \in U$, therefore $(\lambda v) + U = (\lambda w) + U$.

We have identity of addition $0 + U$ and inverse of addition $(-v) + U$ for all $v \in V$. □

Definition 3.104. Let $U \subseteq V$ a subspace, the quotient map $\pi : V \rightarrow V/U$ is a linear mapping defined by:

$$\pi(v) = v + U$$

Proof. We will show π is a linear mapping, $\pi(v + w) = (v + w) + U = v + U + w + U = \pi(v) + \pi(w)$ and $\lambda\pi(v) = \lambda(v + U) = (\lambda v) + U = \pi(\lambda v)$. □

Theorem 3.105. Let V finite and $U \subseteq V$ a subspace, show that $\dim(V/U) = \dim V - \dim U$.

Proof. We can rewrite the equation as $\dim V = \dim(V/U) + \dim U$, and it is easy to see that $\text{range } \pi = \dim(V/U)$ and $\text{null } \pi = \dim U$. $\square$

Definition 3.106. Let $T \in \mathcal{L}(V, W)$, define $\tilde{T} : V/(\text{null } T) \rightarrow W$ by $\tilde{T}(v + \text{null } T) = Tv$.

Theorem 3.107. Let $T \in \mathcal{L}(V, W)$, then:

1. $\tilde{T} \circ \pi = T$
2. $\tilde{T}$ is injective
3. $\text{range } \tilde{T} = \text{range } T$
4. $V/(\text{null } T) \cong \text{range } T$

Proof.

1. For all $v \in V$, $\tilde{T}(\pi(v)) = \tilde{T}(v + \text{null } T) = Tv$
2. If $\tilde{T}(v + \text{null } T) = \tilde{T}(w + \text{null } T)$, then $T(v - w) = 0$, which means $v - w \in \text{null } T$, therefore $v + \text{null } T = w + \text{null } T$.
3. For any $Tv \in \text{range } T$, we have $\tilde{T}(v + \text{null } T) \in \text{range } \tilde{T}$. For any $\tilde{T}(v + \text{null } T) = Tv \in \text{range } \tilde{T}$, we have $Tv \in \text{range } T$.
4. Restrict the range of $\tilde{T}$ on $\text{range } T$, say $\varphi(v + \text{null } T) = \tilde{T}(v + \text{null } T) : V/(\text{null } T) \rightarrow \text{range } T$, then φ is injective since (2) and surjective since (3), therefore φ is an isomorphism, thus $V/(\text{null } T) \simeq \text{range } T$.

$\square$

Definition 3.110 (Dual Space). Let V a vector space, then we denote V' the dual space of V , where

$$V' = \mathcal{L}(V, F)$$

Theorem 3.111. Let V a finite vector space, then $\dim V' = \dim V$

Proof. $\dim V' = \dim \mathcal{L}(V, F) = (\dim V)(\dim F) = \dim V$ $\square$

Definition 3.112 (Dual Basis). Let $v_0, \dots, v_{m-1}$ a basis of V , then the dual basis of $v_0, \dots, v_{m-1}$ is $\varphi_0, \dots, \varphi_{m-1}$ such that:

$$\varphi_i(v_j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

holes for any $0 \leq i, j < m$.

We can see that the basis the dual basis extracts the coefficients of any vector in V .

Theorem 3.113. *Let $v_0, \dots, v_{m-1}$ a basis of V , and dual basis $\varphi_0, \dots, \varphi_{m-1}$ of which. Then for any $v \in V$,*

$$v = \varphi_0(v)v_0 + \dots + \varphi_{m-1}(v)v_{m-1}$$

Proof. For any i , $\varphi_i(v) = \varphi_i(\lambda_0 v_0 + \dots + \lambda_{m-1} v_{m-1}) = \varphi_i(\lambda_i v_i) = \lambda_i \varphi_i(v_i) = \lambda_i \times 1$. $\square$

Theorem 3.116. *Let V a finite space, then the dual basis of basis of V is a basis of V' .*

Proof. Let $v_0, \dots, v_{m-1}$ a basis of V , then its dual basis has the same length, therefore we only need to show its dual basis is linear independent.

Suppose $\lambda_0 \varphi_0 + \dots + \lambda_{m-1} \varphi_{m-1} = 0$, then for any $0 \leq i < m$, $(\lambda_0 \varphi_0 + \dots + \lambda_{m-1} \varphi_{m-1})(v_i) = \lambda_i = 0$, therefore the dual basis is linear independent. $\square$

Definition 3.118 (Dual Map). *Let $T \in \mathcal{L}(V, W)$. A dual map of T is a linear map $T' \in \mathcal{L}(W', V')$, such that for any $\varphi \in W'$:*

$$T'(\varphi) = \varphi \circ T$$

Theorem 3.128. *Let V, W are finite spaces and $T \in \mathcal{L}(V, W)$. Show that:*

1. $\text{null } T' = (\text{range } T)^0$
2. $\dim \text{null } T' = \dim \text{null } T + \dim W - \dim V$

Proof.

- $\text{null } T'$ is a space $\{ \varphi \in \mathcal{L}(W, F) \mid \varphi \circ T = 0 \}$, which means $\text{range } T \subseteq \text{null } \varphi$. $(\text{range } T)^0$ is a space $\{ \varphi \in \mathcal{L}(W, F) \mid \varphi(\text{range } T) = \{0\} \}$, which means $\text{range } T \subseteq \text{null } \varphi$. Therefore $\text{null } T' = (\text{range } T)^0$
- $\dim(\text{range } T)^0 = \dim W - \dim \text{range } T = \dim W - (\dim V - \dim \text{null } T)$

$\square$

Theorem 3.129. *Let V, W are finite spaces and $T \in \mathcal{L}(V, W)$. Show that*

$$T \text{ is surjective} \iff T' \text{ is injective}$$

Proof.

- Suppose T is surjective, then for any $T'(\varphi) = T'(\psi)$, we have $\varphi \circ T = \psi \circ T$. Since T is surjective, then T is an epimorphism (we proved this in E3B), therefore $\varphi = \psi$.
- Suppose T' is injective, then for any $\varphi, \psi \in \mathcal{L}(W, F)$ such that $\varphi \circ T = \psi \circ T$, we have $\varphi = \psi$ since T' is injective. Therefore T is epimorphism, thus surjective.

□

The last theorem is obviously true in category theory, but we haven't show that T' is a morphism in Vect' where $\text{Vect}' \simeq \text{Vect}^{\text{op}}$.

Theorem 3.130. *Let V, W are finite and $T \in \mathcal{L}(V, W)$, show that:*

1. $\dim \text{range } T' = \dim \text{range } T$
2. $\text{range } T' = (\text{null } T)^0$

Proof.

- $\dim \text{range } T' = \dim W' - \dim \text{null } T' = \dim W' - (\dim \text{null } T + \dim W - \dim V) = \dim V - \dim \text{null } T = \dim \text{range } T$
- For any $\varphi \circ T \in \text{range } T'$, $(\varphi \circ T)(\text{null } T) = \varphi(\{0\}) = \{0\}$, therefore $\text{range } T' \subseteq (\text{null } T)^0$. Since $\dim(\text{null } T)^0 = \dim V - \dim \text{null } T = \dim \text{range } T = \dim \text{range } T'$, therefore $\text{range } T' = (\text{null } T)^0$ since both of them are finite and $\text{range } T' \subseteq (\text{null } T)^0$.

□

Theorem 3.131. *Let V, W are finite spaces and $T \in \mathcal{L}(V, W)$. Show*

$$T \text{ is injective} \iff T' \text{ is surjective}$$

Proof.

- $\dim \text{range } T' = \dim \text{range } T = \dim V = \dim V'$ since T is injective, therefore T' is surjective.
- $\dim \text{range } T = \dim \text{range } T' = \dim V' = \dim V$ therefore T is injective.

□

This chapter is much like a note, I will record some idea about polynomial and linear algebra.

One relationship is that a polynomial is a linear combination of the standard basis of $\mathcal{P}(F)$, that is, $1, x, x^2, \dots$. This is important when we apply p to an operator of a vector space, say $T \in \mathcal{L}(V)$ and $p(T) = c_0I + c_1T + c_2T^2 + \dots$. If we apply $p(T)$ to some $v \in V$, it becomes a linear combination of $v, Tv, T^2v, \dots$.

Theorem 3.16. *Let $p \in \mathcal{P}(\mathbb{R})$ is not constant, then p can be factorized into:*

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1x + c_1) \cdots (x^2 + b_Mx + c_M)$$

where $c, \lambda_1, \dots, \lambda_m, b_1, \dots, b_M, c_1, \dots, c_M \in \mathbb{R}$ and for any $1 \leq k \leq M$, $b_k^2 < 4c_k$.

I won't paste the proof here, but the statement can be considered as: any non-constant, real p can be factorized into:

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x - \lambda_{m+1}) \cdots (x - \lambda_{m+M})$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{R}$ and $\lambda_{m+1}, \dots, \lambda_{m+M} \in \mathbb{C}$. Those λ are zeros of p , however some are real, some are complex. This re-expression makes the statement more understandable. Note that λ_{m+k} is paired, since both λ_{m+k} and $\bar{\lambda}_{m+k}$ are zeros of p .

Theorem 3.39. *Let $T \in \mathcal{L}(V)$ and $v_0, \dots, v_{n-1}$ a basis of V , then the following statements are equivalent to each others.*

- $\mathcal{M}(T)$ about $v_0, \dots, v_{n-1}$ is upper-triangular matrix
- For any $k = 1, \dots, n$, $\text{span}(v_0, \dots, v_{k-1})$ is invariant under T .
- For any $k = 1, \dots, n$, $Tv_{k-1} \in \text{span}(v_0, \dots, v_{k-1})$.

Proof.

- (1) $\Rightarrow$ (2) Induction on k . In brief, first k columns are in $\text{span}(v_0, \dots, v_{k-1})$, therefore $\text{span}(v_0, \dots, v_{k-1})$ is invariant under T .
- (2) $\Rightarrow$ (3) Trivial, $v_{k-1} \in \text{span}(v_0, \dots, v_{k-1})$ which is invariant under T .

- (3) $\Rightarrow$ (1) Basically the definition of upper-triangular matrix, $Tv_{k-1} \in \text{span}(v_0, \dots, v_{k-1})$ means the k -th column of $\mathcal{M}(T)$ consists of first k number (the coefficients of Tv_{k-1}) and 0s.

□

Theorem 3.40. *Let $T \in \mathcal{L}(V)$ and $v_0, \dots, v_{n-1}$ a basis of V , such that $\mathcal{M}(T)$ is upper-triangular matrix, and $\lambda_0, \dots, \lambda_{n-1}$ are the numbers of its diagonal. Show that*

$$(T - \lambda_0 I) \cdots (T - \lambda_{n-1} I) = 0$$

Proof. All numbers since i of $(T - \lambda_i I)v$ are 0 if numbers after i of v are 0. Thus $(T - \lambda_0 I) \cdots (T - \lambda_{n-1} I)$ makes $n - 1$ -th number 0, then $n - 2$ -th number and so on. □

Theorem 3.41. *Let $T \in \mathcal{L}(V)$ and $\mathcal{M}(T)$ about some basis of V is upper-triangular matrix. Show that the eigenvalues of T are the numbers in the diagonal.*

Proof. Let $v_0, \dots, v_{n-1}$ a basis of V and $\mathcal{M}(T)$ about this basis is upper-triangular matrix. For λ_i where $i = 0, \dots, n - 1$, we will show that $T - \lambda_i I$ is not invertible.

We will see first i columns of $T - \lambda_i I$ is linear dependent, since they have at most $i - 1$ non-zero numbers while the list they form has length i .

Another part of proof follows the book. Let $q(z) = (z - \lambda_0) \cdots (z - \lambda_{n-1})$, then $q(T) = 0$ by 3.40, thus q is polynomial multiple of the minimal polynomial of T , thus any zero of the minimal polynomial of T is also a zero of q , which means it belongs to the list $\lambda_0, \dots, \lambda_{n-1}$. □

Theorem 3.44. *Let V finite and $T \in \mathcal{L}(V)$. Show that $\mathcal{M}(T)$ is upper-triangular matrix about some basis of $V \iff$ the minimal polynomial of T is in form of $(z - \lambda_0) \cdots (z - \lambda_{n-1})$ where $\lambda_i \in F$.*

Proof. This proof comes from the book.

The $(\Rightarrow)$ part follows theorem 3.41, $(z - \lambda_0) \cdots (z - \lambda_{m-1})$ ($\lambda_0, \dots, \lambda_{m-1}$ are the numbers in the diagonal) is polynomial multiple of the minimal polynomial of T , then the minimal polynomial of T must in a similar form.

For $(\Leftarrow)$, we will induction on n .

- Base($n = 0$), the minimal polynomial of T is in form $(z - \lambda_0)$, thus $T = \lambda_0 I$.

- Ind($n = n + 1$), the minimal polynomial of T is in form $p(z) = (z - \lambda_0) \cdots (z - \lambda_{n+1-1})$. Consider $T - \lambda_n I$, there is non-zero $v \in V$ such that $(T - \lambda_n I)v = 0$ since λ_n is a zero of p therefore an eigenvalue of T . We define $U = \text{range}(T - \lambda_n I)$, consider $q(z) = (z - \lambda_0) \cdots (z - \lambda_{n-1})$ and then $q(T|_U) = 0$, recall that $U = \text{range}(T - \lambda_n I)$, therefore for any $v \in U$, there is u such that $(T - \lambda_n I)u = v$. Thus $q(T|_U)u = q(T)(T - \lambda_n I)v = p(T)v = 0$ where $u = (T - \lambda_n I)v$ and $u, v \in U$. Thus the matrix of $T|_U$ is upper-triangular.

Then we consider $u_0, \dots, u_{k-1}$ a basis of U , we will expand $u_0, \dots, u_{k-1}$ to a basis of V , say $u_0, \dots, u_{k-1}, v_0, \dots, v_{m-1}$. Then for any v_i where i , we have $Tv_i = Tv_i - \lambda_n v_i + \lambda_n v_i = (T - \lambda_n I)v_i + \lambda_n v_i$, where $(T - \lambda_n I)v_i \in U$ (recall the definition of U), thus $Tv_i \in \text{span}(u_0, \dots, u_{k-1}, v_0, \dots, v_i)$, then by 3.39, we know $\mathcal{M}(T)$ is an upper-triangular matrix about the basis $u_0, \dots, u_{k-1}, v_0, \dots, v_{m-1}$.

One may confused that the "length" of u_i is greater than the size of $\mathcal{M}(T|_U) = \dim U$ (thus it won't be a square matrix but tall and thin), however, these two things are unrelated, a matrix only represents how to combine the basis, and doesn't care what the basis looks like.

□

Theorem 3.62. *Let V finite, $T \in \mathcal{L}(V)$. Show that T is diagonalizable $\iff$ the minimal polynomial of T is in form of $(z - \lambda_0) \cdots (z - \lambda_{n-1})$ where $\lambda_0, \dots, \lambda_{n-1} \in F$ are distinct to each others.*

Proof.

- ($\Rightarrow$) Let $p(z) = (z - \lambda_0) \cdots (z - \lambda_{n-1})$ where $\lambda_0, \dots, \lambda_{n-1}$ are distinct numbers in the diagonal of $\mathcal{M}(T)$. Then $p(T)v_i = (T - \lambda_0) \cdots (T - \lambda_i)(T - \lambda_{i+1}) \cdots (T - \lambda_{n-1})v_i$, we can see $(T - \lambda_k)v = (\lambda_i - \lambda_k)v \neq 0$, therefore $(T - \lambda_i)(\lambda v) = 0$ where $\lambda v = (T - \lambda_{i+1}) \cdots (T - \lambda_{n-1})v_i$. In breif, $T - \lambda_k$ eliminates all vector with eigenvalue λ_k in basis in vector $v = \lambda_0 v_0 + \cdots + \lambda_{n-1} v_{n-1}$. therefore $p(T)$ eliminates all v_k of a vector $v \in V$, thus $p(T) = 0$

$$p(T)v = p(T)(\lambda_0 v_0 + \cdots + \lambda_{n-1} v_{n-1}) = \lambda_0 p(T)v_0 + \cdots + \lambda_{n-1} p(T)v_{n-1} = 0 + \cdots + 0 = 0.$$

- ($\Leftarrow$) Induction on n .

Base($n = 1$): $p(T) = (T - \lambda_0 I)$, then $T = \lambda_0 I$, which is diagonal matrix.

Ind($n = n + 1$): Consider $T - \lambda_{n+1} I$ and define $U = \text{range}(T - \lambda_n)$. It is easy to verify that U is invariant under T . Then $q(T|_U) = (T - \lambda_0) \cdots (T - \lambda_{n-1}) = 0$, therefore U is diagonalizable. Thus $U = E(\lambda_0, T|_U) \oplus \cdots \oplus E(\lambda_{n-1}, T|_U)$, note that $\dim V = \dim \text{null } T + \dim \text{range } T = \dim E(\lambda_n, T) + \dim U$, thus we need to show that $U + E(\lambda_n, T)$ is a direct sum, which is immediately holds by the property of eigenspace. Thus $V = U \oplus E(\lambda_n, T)$, which is a direct sum of eigenspace while $\lambda_0, \dots, \lambda_n$ are all distinct eigenvalue of T (since they are zeros of the minimal polynomial of T).

□

Theorem 3.75. *Let $S, T \in \mathcal{L}(V)$ commute and $\lambda \in F$. Show that $E(\lambda, S)$ is invariant under T .*

Proof. For any $v \in E(\lambda, S)$, $STv = TSv = T\lambda v = \lambda Tv$, thus $Tv \in E(\lambda, S)$.

□

Definition 6.2. *A **inner product** of a vector space V is a function that maps $u, v \in V$ to $\langle u, v \rangle \in F$, and it satisfies:*

- *Positivity:* $\langle v, v \rangle \geq 0$.
- *Definiteness:* $\langle v, v \rangle = 0 \iff v = 0$.
- *Additivity:* $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
- *Homogeneity:* $\langle \lambda v, w \rangle = \lambda \langle v, w \rangle$.
- *Conjugate Symmetry:* $\langle u, v \rangle = \overline{\langle v, u \rangle}$

Definition 6.4. *A vector space equipped with an inner product is called an **inner product space**.*

We assume vector spaces V, W are inner product space for the rest of chapter.

Theorem 6.6. *Properties of inner product:*

- *Let $v \in V$, then $\langle -, v \rangle$ is a linear map $V \rightarrow F$.*

- For any $v \in V$, we have $\langle 0, v \rangle = \langle v, 0 \rangle = 0$.
- For any $u, v, w \in V$, $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$.
- For any $v, w \in V$ and $\lambda \in F$, $\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle$.

Proof.

- Trivial from the definition.
- First, $\langle 0, v \rangle = 0$ since $\langle -, v \rangle$ is a linear map, thus maps 0 to 0. Then $\langle v, 0 \rangle = \overline{\langle 0, v \rangle} = \bar{0} = 0$.
- $\langle u, v + w \rangle = \overline{\langle v + w, u \rangle} = \overline{\langle v, u \rangle + \langle w, u \rangle} = \overline{\langle v, u \rangle} + \overline{\langle w, u \rangle} = \langle u, v \rangle + \langle u, w \rangle$.
- $\langle u, \lambda v \rangle = \overline{\langle \lambda v, u \rangle} = \overline{\lambda \langle v, u \rangle} = \bar{\lambda} \overline{\langle v, u \rangle} = \bar{\lambda} \langle u, v \rangle$

□

Definition 6.7. For any $v \in V$, the **norm** of v is denoted by $\|v\|$, and is defined by:

$$\|v\| = \sqrt{\langle v, v \rangle}$$

Theorem 6.9. Let $v \in V$,

- $\|v\| = 0 \iff v = 0$.
- For any $\lambda \in F$, $\|\lambda v\| = |\lambda| \|v\|$.

Proof.

- Trivial by the definition of inner product.
- $\|\lambda v\| = \sqrt{\langle \lambda v, \lambda v \rangle} = \sqrt{\lambda \bar{\lambda} \langle v, v \rangle} = |\lambda| \|v\|$.

□

Definition 6.10. Let $u, v \in V$, u and v are **orthogonal** $\iff \langle u, v \rangle = 0$

Theorem 6.11.

- 0 is orthogonal to any $v \in V$.

- 0 is the only vector that orthogonal to itself.

Proof. Both trivial by the definition, (1) is equivalent to $\langle 0, v \rangle = 0$ and (2) is equivalent to $\langle v, v \rangle = 0 \iff v = 0$. $\square$

Theorem 6.12 (勾股定理). *Let $u, v \in V$, if u is orthogonal to v , then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.*

Proof.

$$\begin{aligned}\|u + w\|^2 &= \langle u + w, u + w \rangle \\ &= \langle u, u + w \rangle + \langle w, u + w \rangle \\ &= \langle u, u \rangle + \langle u, w \rangle + \langle w, u \rangle + \langle w, w \rangle \\ &= \langle u, u \rangle + \langle w, w \rangle\end{aligned}$$

The last equation is by $\langle u, w \rangle = \langle w, u \rangle = 0$ cause u, w are orthogonal. $\square$

Theorem 6.13 (One Orthogonal Factorization). *Let $u, v \in V$ and $v \neq 0$. Let $c = \frac{\langle u, v \rangle}{\|v\|^2}$ and $w = u - cv$, then $u = cv + w$ and $w \perp v$.*

Proof. 嗯算。 $\square$

Theorem 6.32.

Exercise 6.1. *Suppose V is a finite vector space, Show that the only two ideal of $\mathcal{L}(V)$ is $\{0\}$ and $\mathcal{L}(V)$. A subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called an ideal, if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for any $T \in \mathcal{L}(V)$ and $E \in \mathcal{E}$.*

Proof. We will use the concept Matrix. Suppose $\lambda_0 v_0 + \dots + \lambda_n v_n$ the basis of V . We want to construct T_i that $T(\lambda_0 v_0 + \dots + \lambda_n v_n) = \lambda_i v_i$ for all $0 \leq i < n$, which is a matrix with all zero but 1 at i, i .

For any matrix, we can always select a non-zero value at a, b and place it at i, b , this can be done by left multiply a matrix with 1 at i, a (this produce a vector at line i with values from line a), then right multiply a matrix with 1 at i, b (this produce a vector at column b with values from line i).

Also, we can always select a non-zero value at a, b and place it at a, i , this can be done by right multiply a matrix with 1 at b, i , then left multiply a matrix with 1 at a, i .

By combining these two operations, we can select a non-zero value at a, b and place it at i, i . Now, consider any non-zero $E \in \mathcal{E}$, we can construct a

matrix with non-zero value at i, i for every $0 \leq i < \dim V$. These matrices are in $\mathcal{E}$ since $\mathcal{E}$ is an ideal, then we can multiply an appropriate scalar to them so that they are matrices with 1 at i, i . By adding up these matrices, we get I , we know $I \in \mathcal{E}$ since $\mathcal{E}$ is a vector space, and now all $T \in \mathcal{L}(V)$ is also in $\mathcal{E}$ since $\mathcal{E}$ is an ideal, then $\mathcal{E} = \mathcal{L}(V)$.

The only exception is $\mathcal{E} = \{0\}$, in this case we can't pick any non-zero element.

Another solution, hope this one is more simple.

Suppose $\mathcal{E}$ an ideal of $\mathcal{L}(V)$ and non-zero, non-surjective $E \in \mathcal{E}$. Let $v_0, \dots, v_{k-1}$ a basis of $\text{null } E$ and $v_k, \dots, v_{k+n}$ such that Ev_{k+i} is a basis of $\text{range } E$, then we have $n \neq 0$ and $k \neq 0$.

Define A a linear transformation which maps v_i to v_{k+i} for $0 \leq i < \min\{k, n\}$ and maps others to 0, then $\dim \text{range } EA = \min\{k, n\}$.

Expand the basis $w_i = Ev_{k+i}$ of $\text{range } E$ to a basis of V , say $w_0, \dots, w_{m-1}$, define B maps Ev_{k+i} to $w_{\min\{k, n\}+i}$, we always have enough $w_{\min\{k, n\}+i}$ since $m-1 = \dim V = \dim \text{null } E + \dim \text{range } E$ while $\min k, n \leq \dim \text{null } E$, then $\dim \text{range } BE = \dim \text{range } E$ since we just re-map the range E .

Now consider $S = EA + BE$, we have $Sv_i = EAv_i = Ev_{k+i} = w_i \in \text{range } E$ for all $0 \leq i < \min\{k, n\}$ and $Sv_{\min\{k, n\}+i} = BEv_{\min\{k, n\}+i} = w_{\min\{k, n\}+i} \in \text{range } BE$ for all $0 \leq i < \dim \text{range } E$. We can see $\text{range } EA \cap \text{range } BE = \{0\}$ and $\dim \text{range}(EA + BE) = \dim \text{range } E + \min\{k, n\}$, where $k = \dim \text{null } E$ and $n = \dim \text{range } E$, the range of $EA + BE$ gets larger and $EA + BE \in \mathcal{E}$ since $EA, BE \in \mathcal{E}$, if $k > n$ (this is the only case that $EA + BE$ is not surjective), then we continue this process with $E = EA + BE$, the procedure will finally terminate since $\mathcal{L}(V)$ is finite (cause V is finite).

Now we show that any $\mathcal{E}$ with non-zero, non-surjective $E \in \mathcal{E}$ implies a surjective (thus injective and invertible) $T \in \mathcal{E}$.

For any ideal with an invertible element $E \in \mathcal{E}$, we have $E^{-1}E = I \in \mathcal{E}$, which causes $\mathcal{E} = \mathcal{L}(V)$ since $IT = T$ for all $T \in \mathcal{L}(V)$.

Therefore, only $\{0\}$ and $\mathcal{L}(V)$ are ideals of $\mathcal{L}(V)$. $\square$

Exercise 6.7. Suppose vector space V and W are finite ($2 \leq \dim V \leq \dim W$), show that $\{ T \in \mathcal{L}(V, W) \mid T \text{ is not injective} \}$ is not a subspace.

Proof. Consider the basis $v_0 + \dots + v_{(\dim V - 1)}$ in V , and $T(v_0 + \dots + v_{(\dim V - 1)}) = (0 + 1_0 + \dots + 1_{(\dim V - 1)})$ and $T'(v_0 + \dots + v_{(\dim V - 1)}) = (v_0 + 0 + \dots + v_{(\dim V - 1)})$. Then $(T + T')(\lambda_0 v_0 + \lambda_1 v_1 + \dots + \lambda_{(\dim V - 1)} v_{(\dim V - 1)}) = \lambda_0 v_0 + \lambda_1 v_1 + \dots + 2\lambda_{(\dim V - 1)} v_{(\dim V - 1)}$, which is obviously injective. $\square$

Exercise 6.11. Suppose V is finite and $T \in \mathcal{L}(V, W)$, show that there is a subspace $U \subset V$ such that:

$$U \cap \text{null } T = \{0\} \quad \text{and} \quad \text{range } T = \{ Tu \mid u \in U \}$$

Proof. This is similar to the *isomorphism theorems* about groups! This can be done by the similar way we used in proving $\dim V = \dim \text{null } T + \dim \text{range } T$. $\square$

The next two exercises remind me the categorical injective and surjective, let try them first!

Exercise. For any $F \in \mathcal{L}(V, W)$, F is injective $\iff$ for any $S, T \in \mathcal{L}(U, V)$, $FS = FT$ implies $S = T$.

Proof.

- $(\Rightarrow)$ For any $S, T \in \mathcal{L}(V, W)$ that $FS = FT$, then for any $u \in U$, we have $F(Su) = F(Tu)$, since F is injective, we know $Su = Tu$, so $S = T$.
- $(\Leftarrow)$ For any $v, w \in V$ such that $Fv = Fw$. Consider

$$\begin{aligned} S(\lambda) &= \lambda v \\ T(\lambda) &= \lambda w \end{aligned}$$

in $\mathcal{L}(\mathbb{R}, V)$. Then for any $\lambda \in \mathbb{R}$, we have $FS\lambda = \lambda FS1 = \lambda Fv = \lambda Fw = \lambda FT1 = FT\lambda$. so $FS = FT$ then $S = T$, which means $v = S1 = T1 = w$. $\square$

Exercise. Suppose W is finite, then for any $F \in \mathcal{L}(V, W)$, F is surjective $\iff$ for any $S, T \in \mathcal{L}(W, U)$, $SF = TF$ implies $S = T$.

Proof.

- $(\Rightarrow)$ For any $S, T \in \mathcal{L}(W, U)$ such that $SF = TF$. For any $w \in W$, there is $v \in V$ such that $Fv = w$ since F is surjective. Then we have $SFv = TFv$ so $Sw = S(Fv) = T(Fv) = Tw$ then $S = T$.

- ($\Leftarrow$) Consider

$$S = I \quad \text{and} \quad T(\lambda_0 w_0 + \cdots + \lambda_n w_n) = \lambda_0 w_0 + \cdots + \lambda_n w_n$$

where $w_0, \dots, w_k$ is the basis of $\text{range } F$ and $w_0, \dots, w_n$ is the basis of W that expand from $w_0, \dots, w_k$.

(If we can use another way to construct T , then W is not need to be finite, for example, $W = \text{range } T \oplus W_0$).

It is easy to show that T is a linear transformation. Then for any $v \in V$, we have $TFv = Fv$ (since T acts like identity transformation on $\text{range } F$) and $SFv = Fv$, so $S = T$ by the property of F . Since $\text{range } S = W$, so is $\text{range } T$, that means $w_0, \dots, w_k$ spans W , so $k = n$, which means $\text{range } F = W$, therefore F is surjective.

□

Exercise 6.19. Suppose W is finite, then for any $T \in \mathcal{L}(V, W)$, show that T is injective $\iff$ there is $S \in \mathcal{L}(W, V)$ such that $ST = I$.

Proof.

- ($\Rightarrow$) Consider the basis $v_0, \dots, v_n$ of V , then $Tv_0, \dots, Tv_n$ is a basis of $\text{range } T$ since T is injective. We denote Tv_i as w_i and $w_0, \dots, w_m$ as the basis of W which expand from $w_0, \dots, w_n$. Define $S(\lambda_0 w_0 + \cdots + \lambda_m w_m) = \lambda_0 w_0 + \cdots + \lambda_n w_n$, and then for any $v \in V$, $ST(\lambda_0 v_0 + \cdots + \lambda_n v_n) = S(\lambda_0 w_0 + \cdots + \lambda_n w_n) = \lambda_0 v_0 + \cdots + \lambda_n v_n$, so $ST = I$.
- ($\Leftarrow$) Suppose $A, B \in \mathcal{L}(U, V)$, such that $TA = TB$, we will show that $A = B$. $STA = IA = A$ and $STB = IB = B$ and $STA = STB$ since $TA = TB$. Then we know T is a monomorphism, and then T is injective.

□

Exercise 6.20. Suppose W is finite, then for any $T \in \mathcal{L}(V, W)$, show that T is surjective $\iff$ there is $S \in \mathcal{L}(W, V)$ such that $TS = I$.

Proof.

□

Exercise 6.21. Suppose V is finite, $T \in \mathcal{L}(V, W)$, $U \subseteq W$ a subspace. Show that the inverse image of U : $\{ v \in V \mid Tv \in U \}$ is a subspace of V , and

$$\dim\{ v \in V \mid Tv \in U \} = \dim \text{null } T + \dim(U \cap \text{range } T)$$

Proof. The second part is quite easy, we can restrict the domain of T to $\{ v \in V \mid Tv \in U \}$, say $T' \in \mathcal{L}(\{ v \in V \mid Tv \in U \}, W)$, so that it is in form $\dim\{ v \in V \mid Tv \in U \} = \dim \text{null } T' + \dim \text{range } T'$. Obviously $\text{range } T' = U \cap \text{range } T$ and $\text{null } T' = \text{null } T$.

We will now show that $\{ v \in V \mid Tv \in U \}$ is a subspace of V .

- $T0 \in U$.
- For any $v, w \in V$ such that $Tv, Tw \in U$, we have $T(v+w) = Tv + Tw \in U$.
- For any $v \in V$ such that $Tv \in U$ and $\lambda \in F$, we have $T(\lambda v) = \lambda Tv \in U$.

Therefore it is a subspace. □

Exercise 6.22. Suppose U and V are finite, $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$, show that

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T$$

Proof. Consider the inverse image of $\text{null } S$ on T : $K = \{ v \in V \mid Tv \in \text{null } S \}$, which dimension: $\dim K = \dim \text{null } T + \dim(\text{null } S \cap \text{range } T)$, where $\dim(\text{null } S \cap \text{range } T)$ caps at $\dim \text{null } S$.

We know show that $\text{null } ST = \text{null } K$. For any $STv = 0$, we know $S(Tv) = 0$, so $Tv \in K$, therefore $\text{null } ST \subseteq \text{null } K$; For any $Tv \in \text{null } S$, that means $S(Tv) = 0$, therefore $v \in \text{null } ST$, therefore $\text{null } ST \supseteq \text{null } K$, and $\text{null } ST = \text{null } K$. □

Exercise 6.25. Suppose V is finite, $S, T \in \mathcal{L}(V, W)$. Show that $\text{null } S \subseteq \text{null } T \iff$ there is $E \in \mathcal{L}(W)$ such that $T = ES$.

Proof. We define $E(S(v)) = Tv$ for any $v \in V$, so that $E \in \mathcal{L}(\text{range } S, W)$. We first show that E is a mapping, and also a linear transformation.

Suppose $Sv, Sw \in W$ such that $Sv = Sw$, we need to show that $E(Sv) = E(Sw)$, or normalized $Tv = Tw$. We know $v - w \in \text{null } S$ since $Sv = Sw$, so $v - w \in \text{null } T$ since $\text{null } S \subseteq \text{null } T$, therefore $T(v - w) = 0$, and then $Tv = Tw$, so E is a mapping.

Now we show that E is a linear transformation.

- For any $Sv, Sw \in \text{range } S$, $E(Sv) + E(Sw) = Tv + Tw = T(v + w) = E(S(v + w)) = E(Sv + Sw)$.

- For any $Sv \in \text{range } S$ and $\lambda \in F$, $\lambda E(Sv) = \lambda Tv = T(\lambda v) = E(S(\lambda v)) = E(\lambda Sv)$.

therefore E is a linear transformation.

Now we can expand the domain of E to W such that $E'v = Ev$ for any $v \in \text{range } S$ (this is proven in previous exercise). For any $v \in V$, we have $ESv = E(Sv) = Tv$, therefore $T = ES$.

For another direction, for any $v \in \text{null } S$, we have $ESv = E0 = 0 = Tv$, so $v \in \text{null } T$. $\square$

Exercise 6.26. Suppose V is finite, $S, T \in \mathcal{L}(V, W)$. Show that $\text{range } S \subseteq \text{range } T \iff$ there is $E \in \mathcal{L}(V)$ such that $S = TE$.

Proof. Consider the inverse image of $\text{range } S$ with basis $w_0, \dots, w_n$, say $v_0, \dots, v_n$, it is easy to show $v_0, \dots, v_n$ is linear independent. Then $E(v) = \lambda_0 v_0 + \dots + \lambda_n v_n$ where $Sv = \lambda_0 w_0 + \dots + \lambda_n w_n$. $\square$

Exercise 6.27. Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$, show that $V = \text{null } P \oplus \text{range } P$.

Proof. Such element is called *idempotent* in algebra.

We will show $\text{null } P \oplus \text{range } P$ by showing $\text{null } P \cap \text{range } P = \{0\}$. For any $v \in \text{null } P \cap \text{range } P$, we know there is $w \in V$ such that $Pw = v$ since $v \in \text{range } P$, then $P^2(v) = P(Pv) = P0 = 0$ since $v \in \text{null } P$ and $P^2(v) = P(Pv) = Pw$, so $Pw = 0$ while $Pw = v$ therefore $v = 0$.

Then we have $\dim V = \dim \text{null } P + \dim \text{range } P$ and $\dim(\text{null } P \oplus \text{range } P) = \dim \text{null } P + \dim \text{range } P - \dim \{0\}$, so $V = \text{null } P \oplus \text{range } P$. $\square$

Exercise 6.28. Suppose $D \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ such that for any non-constant polynomial $p \in \mathcal{P}(\mathbb{R})$, $\deg(Dp) = \deg p - 1$. Show that D is surjective.

Proof. We induction on n , starts from 1, to show that $D(\mathcal{P}_n(\mathbb{R})) = \mathcal{P}_{n-1}(\mathbb{R})$.

- Base: for any $p \in \mathcal{P}(\mathbb{R})$ where $\deg p = 1$, we know $\deg Dp = 0$, so $D(\mathcal{P}_1(\mathbb{R}))$ is a non-zero subspace of $\mathcal{P}_0(\mathbb{R})$, which is $\mathcal{P}_0(\mathbb{R})$.
- Induction: We have induction hypothesis: For any $i \leq n$, we have $D(\mathcal{P}_i(\mathbb{R})) = \mathcal{P}_{i-1}(\mathbb{R})$. We want to show that $D(\mathcal{P}_{n+1}(\mathbb{R})) = \mathcal{P}_n(\mathbb{R})$. For any $p \in \mathcal{P}(\mathbb{R})$ with $\deg p = n+1$, we can write p in form of $p = \lambda x^{n+1} + r$ where $\deg r \leq n$, then $Dp = D(\lambda x^{n+1} + r) = D(\lambda x^{n+1}) + Dr$ where $\deg D(\lambda x^{n+1}) = n$ and $\deg Dr \leq n-1$. So $D(\mathcal{P}_{n+1}(\mathbb{R})) = \mathcal{P}_n(\mathbb{R})$ since: $\mathcal{P}_n(\mathbb{R}) \subseteq D(\mathcal{P}_{n+1}(\mathbb{R}))$ and $D(\lambda x^{n+1}) \in D(\mathcal{P}_{n+1}(\mathbb{R}))$, it is sufficient to span $\mathcal{P}_n(\mathbb{R})$.

□

Exercise 6.29. For any $p \in \mathcal{P}(\mathbb{R})$, show that there is $q \in \mathcal{P}(\mathbb{R})$ such that $5q'' + 3q' = p$.

Proof. We can rewrite the goal as $5DDq + 3Dq = p$ where $D(p) = p'$, then $5DDq + 3Dq = D(5Dq) + D(3q) = D(5Dq + 3q) = p$. We know D is surjective by the previous exercise, the goal is now showing that $5Dq + 3q = r$ where $Dr = p$. Then we continue rewrite the goal $5Dq + 3q = (5D)q + (3I)q = (5D + 3I)q = r$, we will show that $5D + 3I$ is surjective, we use the same method in previous exercise.

We denote $5D + 3I$ by F , and induction on $n \in \mathbb{N}$ to show that $F(\mathcal{P}_n(\mathbb{R})) = \mathcal{P}_n(\mathbb{R})$.

- Base: We should show that $F(\mathcal{P}_0(\mathbb{R})) = \mathcal{P}_0(\mathbb{R})$, for any $p \in \mathcal{P}_0(\mathbb{R})$, we have $Fp = 5Dp + 3p$, where $Dp = 0$ since $\deg p = 0$, so $Fp = 3p$, which means we have $1 \in F(\mathcal{P}_0(\mathbb{R}))$ since p is literally a number and $\frac{1}{3p}Fp = 1$.
- Induction: We have induction hypothesis: $F(\mathcal{P}_n(\mathbb{R})) = \mathcal{P}_n(\mathbb{R})$, and we want to show $F(\mathcal{P}_{n+1}(\mathbb{R})) = \mathcal{P}_{n+1}(\mathbb{R})$.

For any $p \in \mathcal{P}_{n+1}(\mathbb{R})$, we have $Fp = 5Dp + 3p$ where $\deg 5Dp = n$ and $\deg 3p = n + 1$, then we can eliminate $5Dp$ and every term in p with degree less than $n + 1$ since $\mathcal{P}_n(\mathbb{R}) \subseteq \text{range } F$, then we get z^{n+1} , thus $F(\mathcal{P}_{n+1}(\mathbb{R})) = \mathcal{P}_{n+1}(\mathbb{R})$.

Therefore there is q such that $(5D + 3I)q = r$ since $5D + 3I$ is surjective.

Another solution from internet: Define $Tq = 5q'' + 3q'$, we can see for any $q \in \mathcal{P}(\mathbb{R})$ we have $\deg Tq = \deg q - 1$, so T is surjective. Then there is q such that $Tq = 5q'' + 3q' = p$. □

Exercise 6.30. Suppose $\varphi \in \mathcal{L}(V, F)$ not zero, and $u \in V$ that $u \notin \text{null } \varphi$, show that $V = \text{null } \varphi \oplus \{ au \mid a \in F \}$.

Proof. We can see φ is surjective since $\varphi u \neq 0$, then for any $i \in F$, we have $(i(\varphi u)^{-1})\varphi u = i$.

For any $v \in V$, since φ is surjective (in a particular way), so we have $a\varphi u$ such that $a\varphi u = \varphi v$, then $\varphi(au - v) = 0$ so $au - v \in \text{null } \varphi$. That means $(-1)(au - v) + au = v$ where $(-1)(au - v) \in \text{null } \varphi$ and $au \in \{ au \mid a \in F \}$, so $V = \text{null } \varphi + \{ au \mid a \in F \}$.

Then $\text{null } \varphi \oplus \{ au \mid a \in F \}$ since $u \notin \text{null } \varphi$. □

Exercise 6.31. Suppose V is finite ($\dim V > 1$), show that if $\varphi : \mathcal{L}(V) \rightarrow F$ is a linear mapping with property $\varphi(ST) = \varphi(S)\varphi(T)$ for any $S, T \in \mathcal{L}(V)$, show that $\varphi = 0$.

Proof. Consider $\text{null } \varphi$, since $\dim V > 1$ while $\dim F = 1$, so φ cannot be injective, therefore $\text{null } \varphi \neq \{0\}$.

For any non-zero $S \in \text{null } \varphi$ and $T \in \mathcal{L}(V)$, we have $\varphi(ST) = \varphi(S)\varphi(T) = 0 = \varphi(T)\varphi(S) = \varphi(TS)$ since $S \in \text{null } \varphi$, thus $ST \in \text{null } \varphi$. We show that $\text{null } \varphi$ is an ideal of $\mathcal{L}(V)$, recall that the property of $\mathcal{L}(V)$, the only ideal of $\mathcal{L}(V)$ is $\{0\}$ and $LT(V)$, so $\text{null } \varphi = \mathcal{L}(V)$, which means $\varphi = 0$. $\square$

Exercise 6.32. Let V, W are vector spaces and $T \in \mathcal{L}(V, W)$, define $T_C : V_C \rightarrow W_C$:

$$T_C(u + iv) = Tu + iTv$$

for any $u, v \in V$.

1. Show that T_C is a (complex) linear mapping from V_C to W_C .
2. Show that T_C is injective $\iff T$ is injective.
3. Show that $\text{range } T_C = W_C \iff \text{range } T = W$.

Proof.

1. For any $u, v, s, t \in V$ $\lambda \in \mathbb{C}$, we have

$$\begin{aligned} & T((u + iv) + (s + it)) \\ &= T(u + s + i(v + t)) \\ &= T(u + s) + iT(v + t) \\ &= Tu + Ts + iTv + iTt \\ &= T(u + iv) + T(s + it) \end{aligned}$$

and

$$\begin{aligned} & \lambda T(u + iv) \\ &= \lambda(Tu + iTv) \\ &= \lambda Tu + \lambda iTv \\ &= T(\lambda u) + iT(\lambda v) \\ &= T(\lambda u + i(\lambda v)) \\ &= T(\lambda u + i\lambda v) \\ &= T(\lambda(u + iv)) \end{aligned}$$

I believe these are trivial, so the future me should be able to prove these without any effort. $\square$

Exercise 6.4. Suppose $D(p) = p' : \mathcal{L}(\mathcal{P}_3(\mathbb{R})) \rightarrow \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$. Find a basis of $\mathcal{L}(\mathcal{P}_3(\mathbb{R}))$ and a basis of $\mathcal{L}(\mathcal{P}_2(\mathbb{R}))$, such that $\mathcal{M}(D)$ about these basis is:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Proof. Consider $x, x^2, x^3, 1$ the basis of $\mathcal{P}_3(\mathbb{R})$ and $1, x, 2x^2$. $\square$

Exercise 6.5. Suppose V and W are finite and $T \in \mathcal{L}(V, W)$. Show that there are basis of V and W respectively, such that $\mathcal{M}(T, \text{those basis})$ is all zero except 1 at k, k ($1 \leq k \leq \dim \text{range } T$).

Proof. Consider the basis $w_0, \dots, w_{k-1}$ of $\text{range } T$ and the basis $w_0, \dots, w_{m-1}$ of W which expands from $w_0, \dots, w_{k-1}$. Then there must be $v_0, \dots, v_{k-1}$ such that $Tv_i = w_i$ for all $0 \leq i < k$, we know $v_0, \dots, v_{k-1}$ is linear independent since $w_0, \dots, w_{k-1}$ is linear independent, so we can expand it to a basis of V , say $v_0, \dots, v_{n-1}$.

We claim that $\mathcal{M}(T, v_0, \dots, v_{n-1}, w_0, \dots, w_{m-1})$ is a matrix with all zero but 1 at k, k ($1 \leq k < \text{range } T$). For any $\lambda_0 v_0 + \dots + \lambda_{n-1} v_{n-1} \in V$, we have $T(\lambda_0 v_0 + \dots + \lambda_{n-1} v_{n-1}) = \lambda_0 w_0 + \dots + \lambda_{k-1} w_{k-1}$, note that all v_i where $i \geq k$ disappear, since they maps to 0. Therefore $\mathcal{M}(T)$ is all zero but 1 at k, k (since $\lambda_i w_i$ in the last equation). $\square$

Exercise 6.6. Show that $-^T : F^{m,n} \rightarrow F^{n,m}$ is a linear mapping.

Proof. Trivial, sorry. $\square$

Exercise 6.7. Show that $(AB)^T = B^T A^T$.

Proof. Suppose A is a $m \times n$ matrix and B is a $n \times p$ matrix, then for any $i \in [1, m]$ and $j \in [1, p]$, we have $(AB)^T_{i,j} = (AB)_{j,i} = \sum_{r=1}^n A_{j,r} B_{r,i} =$

$$\sum_{r=1}^n B_{i,r}^T A_{r,j}^T = (B^T A^T)_{i,j}. \quad \square$$

Exercise 6.8. Let A a $m \times n$ matrix, show that the rank of A is 1 $\iff$ there is $c_0, \dots, c_{m-1} \in F^m$ and $d_0, \dots, d_{n-1} \in F^n$ such that $A_{j,k} = c_j d_k$ for all $j = 0, \dots, m-1$ and $k = 0, \dots, n-1$.

Proof. The right hand side is actually the external product of vectors, that is vw^T .

$(\Rightarrow)$ is easy since we can use the theorem that any $m \times n$ matrix A can be expressed by CR where C is a $m \times r$ matrix, R is a $r \times n$ matrix, r is the rank of A . In this case, $r = 1$, so C and R are just vectors.

$(\Leftarrow)$ is also easy since other column is a scalar multiple of the first column, therefore the rank of A is 1. $\square$

Exercise 6.9. Let $T \in \mathcal{L}(V)$, $u_0, \dots, u_{n-1}$ and $v_0, \dots, v_{n-1}$ are the bases of V , show that the following statements are equivalent:

1. T is injective
2. The columns of $\mathcal{M}(T)$ is linear independent
3. The columns of $\mathcal{M}(T)$ spans $F^{n,1}$
4. The lines of $\mathcal{M}(T)$ is linear independent
5. The lines of $\mathcal{M}(T)$ spans $F^{1,n}$

Proof. (2), (3) are obviously equivalent and (4), (5) too.

Although I want to make an arrow loop, but the arrow between (1) and (4), (5) is too hard, so I will show that (1) $\iff$ (2), (3) and (2), (3) $\iff$ (4), (5).

- $(\Rightarrow)$ Let $\lambda_0 w_0 + \dots + \lambda_{n-1} w_{n-1} = [0, \dots, 0]$, then $T(\lambda_0 u_0 + \dots + \lambda_{n-1} u_{n-1}) = 0$, so λ_i are 0 since T is injective, which means $\text{null } T = \{0\}$.
- $(\Leftarrow)$ For any $T(\lambda_0 u_0 + \dots + \lambda_{n-1} u_{n-1}) = 0$, we have the linear combination of v_i is 0 where the coefficients come from $\lambda_0 w_0 + \dots + \lambda_{n-1} w_{n-1}$ (w_i are the columns of $\mathcal{M}(T)$), therefore the coefficients are all 0 since v_i is linear independent, thus $\lambda_0 w_0 + \dots + \lambda_{n-1} w_{n-1} = 0$, which means λ_i are all 0 since w_i is linear independent.
- For any matrix, its line rank is equal to its column rank, so columns independent $\iff$ lines independent.

$\square$

Exercise 6.4. Let V a finite vector space with $\dim V > 1$, show that $S = \{ T \text{ is singular} \mid T \in \mathcal{L}(V) \}$ is **NOT** a subspace of $\mathcal{L}(V)$.

Proof. If S is a subspace of $\mathcal{L}(V)$, then it is an ideal of $\mathcal{L}(V)$ since for any $A \in S$ and $B \in \mathcal{L}(V)$, AB and BA are singular, therefore $AB, BA \in S$. However, we know the only two ideals of $\mathcal{L}(V)$ is $\{0\}$ and $\mathcal{L}(V)$, none of them is S . $\square$

Exercise 6.11. Let V finite vector space, and $S, T \in \mathcal{L}(V)$, show that

$$ST \text{ is invertible} \iff S \text{ and } T \text{ are invertible}$$

Proof.

- ($\Rightarrow$) Suppose $STW = WST = I$, then $S(TW) = (TW)S = I$ since $\dim V = \dim V$, therefore $S^{-1} = TW$, also $(WS)T = T(WS) = I$ since $\dim V = \dim V$, therefore $T^{-1} = WS$.
- ($\Leftarrow$) Trivial.

$\square$

Exercise 6.12. Let V finite vector space, and $S, T, U \in \mathcal{L}(V)$ such that $STU = I$, Show that $T^{-1} = US$.

Proof. Since $STU = I$ we know U is invertible (since STU is invertible), then $ST = U^{-1}$. Since U^{-1} is invertible, we know S and T are invertible therefore $T = S^{-1}U^{-1}$ and $T^{-1} = US$. $\square$

Exercise 6.13. Show that the conclusion of previous exercise can be false if V is not finite.

Proof. Let $S(x_0, x_1, \dots) = (x_1, \dots)$ the backward-shift mapping and $U(x_0, x_1, \dots) = (0, x_0, x_1, \dots)$ the forward-shift mapping and $T = I$ the identity mapping.

We have $SU = I$ and $US \neq I$, T is clearly invertible with $T^{-1} = I$, but we know $US \neq I$, so $T^{-1} = US \neq I$.

In fact, this also disprove the infinite version of exercise 6.11 since SU is invertible but neither S nor U is invertible. $\square$

Exercise 6.17. Let V a finite vector space, $S \in \mathcal{L}(V)$, define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}(T) = ST$, show that:

1. $\dim \text{null } \mathcal{A} = (\dim V)(\dim \text{null } S)$
2. $\dim \text{range } \mathcal{A} = (\dim V)(\dim \text{range } S)$

Proof. Since $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$, we know $\dim \mathcal{L}(V) = \dim \text{null } \mathcal{A} + \dim \text{range } \mathcal{A}$, also, $\dim \mathcal{L}(V) = (\dim V)^2$ and $\dim V = \dim \text{null } S + \dim \text{range } S$. Therefore we have $\dim \text{null } \mathcal{A} + \dim \text{range } \mathcal{A} = (\dim V)(\dim \text{null } S + \dim \text{range } S)$, which means we only need to prove one of (1) and (2).

We will show that $\dim \text{null } \mathcal{A} = (\dim V)(\dim \text{null } S)$. We found that $\dim \mathcal{L}(V, \text{null } S) = (\dim V)(\dim \text{null } S)$, so it would be nice if $\text{null } \mathcal{A} = \mathcal{L}(V, \text{null } S)$. For any $T \in \text{null } \mathcal{A}$, we have $ST = 0$, which means $\text{range } T \subseteq \text{null } S$, therefore $T \in \mathcal{L}(V, \text{null } S)$. For any $T \in \mathcal{L}(V, \text{null } S)$, we have $ST = 0$ since $\text{range } T \subseteq \text{null } S$, so $T \in \text{null } \mathcal{A}$, therefore $\text{null } \mathcal{A} = \mathcal{L}(V, \text{null } S)$, thus $\dim \text{null } \mathcal{A} = (\dim V)(\dim \text{null } S)$. $\square$

Exercise 6.18. Show that V and $\mathcal{L}(F, V)$ are isomorphic.

Proof. This can be proven by $\dim V = \dim \mathcal{L}(F, V) = 1(\dim V)$, but we can find $\varphi(v) = x \mapsto xv$ an isomorphism. For any $T \in \mathcal{L}(F, V)$, T is determined by $T(1)$. $\square$

Exercise 3.1. Let $T : V \rightarrow W$, the graph of T is a subset of $V \times W$ such that

$$\text{graph of } T = \{ (v, Tv) \mid v \in V \}$$

Show that T is a linear mapping $\iff$ the graph of T is a subspace.

Proof.

- $(0, T0) \in \text{graph of } T$. $(v, Tv) + (w, Tw) = (v + w, Tv + Tw) = (v + w, T(v + w))$. $\lambda(v, Tv) = (\lambda v, \lambda Tv) = (\lambda v, T(\lambda v))$
- $(v, Tv) + (w, Tw) = (v + w, T(v + w))$ since the graph of T is a subspace, therefore $Tv + Tw = T(v + w)$. Similarly, $\lambda Tv = T(\lambda v)$.

$\square$

Exercise 3.3. Let V_i are vector spaces, show that $\mathcal{L}(V_0 \times \cdots \times V_{m-1}, W) \simeq \mathcal{L}(V_0, W) \times \cdots \times \mathcal{L}(V_{m-1}, W)$.

Proof. This can be proven by $A \times B$ is a categorical product, so we will show that for any A, B are vector spaces, $A \times B$ is a product.

In order to show that $A \times B$ is a product, or more specifically, $A \times B$ equipped with linear mappings

$$\begin{aligned}\pi_0(a, b) &= a \\ \pi_1(a, b) &= b\end{aligned}$$

is a product, we have to show that for any C , $s \in \mathcal{L}(C, A)$ and $t \in \mathcal{L}(C, B)$, there is a unique $u \in \mathcal{L}(C, A \times B)$ such that $s = \pi_0 \circ u$ and $t = \pi_1 \circ u$.

Define $u(c) = (sc, tc) : C \rightarrow A \times B$, we will show that u is a linear mapping.

- For all $v, w \in C$, $u(v) + u(w) = (sv, tv) + (sw, tw) = (sv + sw, tv + tw) = (s(v + w), t(v + w)) = u(v + w)$
- For all $c \in C$ and $\lambda \in F$, $\lambda u(c) = \lambda(sc, tc) = (\lambda sc, \lambda tc) = (s(\lambda c), t(\lambda c)) = u(\lambda c)$.

Then we can see $\pi_0(u(c)) = \pi_0(sc, tc) = sc$ and $\pi_1(u(c)) = \pi_1(sc, tc) = tc$. Now we have to show that u is unique (which is trivial, I don't want to prove this, sorry). $\square$

Exercise 3.5. Let m a positive number, define $V^m = \underbrace{V \times \cdots \times V}_m$, show that $V^m \simeq \mathcal{L}(F^m, V)$.

Proof. Define $\varphi(v_0, \dots, v_{m-1}) = i_0, \dots, i_{m-1} \mapsto i_0 v_0 + \cdots + i_{m-1} v_{m-1}$ which accept a list of vector and a list of coefficients then produce a linear combination.

For any $T \in \mathcal{L}(F^m, V)$, T is completely determined by $T(1, \dots, 1) = v_0 + \cdots + v_{m-1}$, therefore $\varphi(v_0, \dots, v_{m-1}) = T$ and thus φ is surjective.

For any $(v_0, \dots, v_{m-1}), (w_0, \dots, w_{m-1}) \in V^m$ such that $\varphi(v_0, \dots, v_{m-1}) = \varphi(w_0, \dots, w_{m-1})$, then $w_0 = \varphi(v_0, \dots, v_{m-1})(1, 0, \dots, 0) = \varphi(v_0, \dots, v_{m-1})(1, 0, \dots, 0) = v_0$, same for other v_i and w_i , so $(v_0, \dots, v_{m-1}) = (w_0, \dots, w_{m-1})$, therefore φ is injective. $\square$

Exercise 3.6. Let $v, x \in V$ and $U, W \subseteq V$ are subspaces such that $v + U = x + W$. Show that $U = W$.

Proof. We know $v = x + w_0$ for some $w_0 \in W$ since $v + U = x + W$ and $v \in v + U$, then for any $u \in U$, we have $v + u = x + w$ for some $w \in W$, then $(x + w_0) + u = x + w$ therefore $u = x + w - x - w_0 = w - w_0 \in W$ thus $U \subseteq W$. Similarly $W \subseteq U$. $\square$

Exercise 3.7. Let $U = \{ (x, y, z) \in R^3 \mid 2x + 3y + 5z = 0 \}$ and $A \subseteq R^3$. Show that A is a translate of U (that is $A = a + U$) $\iff$ there is c such that $A = \{ (x, y, z) \in R^3 \mid 2x + 3y + 5z = c \}$.

Proof.

- ($\Rightarrow$) For any $(a_0, a_1, a_2) + (x, y, z) \in a + U$, we have $2(a_0 + x) + 3(a_1 + y) + 5(a_2 + z) = 2a_0 + 3a_1 + 5a_2$, therefore $c = 2a_0 + 3a_1 + 5a_2$.
- ($\Leftarrow$) We can see 2, 3 and 5 are coprime to each other, therefore there is $2a_0 + 3a_1 + 5a_2 = 1$ (I am not sure if this is true in generalized case, I just extends the theorem " $as + bt = 1 \iff a$ coprime to b " to three elements case without checking), in this case we have $2(1) + 3(-2) + 5(1) = 1$, then for any $2x + 3y + 5z = c$, we have $2x + 3y + 5z = 2(ca_0) + 3(ca_1) + 5(ca_2)$, then $2(x - ca_0) + 3(y - ca_1) + 5(z - ca_2) = 0$, therefore $A = ((-c)(a_0, a_1, a_2)) + U$.

□

Exercise 3.8. Let $T \in \mathcal{L}(V, W)$ and $c \in W$, show that $\{ v \in V \mid Tv = c \}$ is an empty set or a translate of $\text{null } T$. Then explain why the solutions of a system of linear equations is either an empty set or a translate of some subspace of F^n .

Proof. Let $Ta = c$ for some $a \in V$, if no such a , then $\{ v \in V \mid Tv = c \} = \emptyset$. We claim $\{ v \in V \mid Tv = c \} = a + \text{null } T$. For any $v \in V$ such that $Tv = c$, then $v = a + v - a$ and $T(v - a) = Tv - Ta = c - c = 0$, therefore $v - a \in \text{null } T$, thus $v \in a + \text{null } T$. In another direction, for any $a + v \in a + \text{null } T$, we have $T(a + v) = Ta + Tv = c + 0 = c$. □

Exercise 3.9. Let $A \subseteq V$ a non-empty subset. Show that A is a translate of some subspace of $V \iff \lambda v + (1 - \lambda)w \in A$ for any $v, w \in A$ and $\lambda \in F$.

Proof.

- ($\Rightarrow$) Suppose $A = a + U$ for some subspace $U \subseteq V$.
- ($\Leftarrow$) Let $w \in A$, we will show that $(-w) + A$ is a subspace of V .

For any $a - w, b - w \in (-w) + A$, we need to show that $a - w + b - w = (a + b - w) - w \in (-w) + A$ or equivalently $a + b - w \in A$. We found that the property $\lambda v + (1 - \lambda)w \in A$ gives us the ability to construct something like $v - w$. Since $2v + (1 - 2)w = 2v - w$, we just let $w = v + a$ then $2v - (v + a) = v - a$. Therefore, we let $\lambda = 2$, $v = a + b$ and $w = a + b + w$, and now $2(a + b) - (a + b + w) = a + b - w \in A$, so $a + b - w - w \in (-w) + A$.

For any $a - w \in (-w) + A$ and $\lambda \in F$, we need to show that $\lambda(a - w) \in (-w) + A$. $\lambda(a - w) = \lambda a - \lambda w = \lambda a - (\lambda - 1)w - w$. We let

$\lambda = (-1)(\lambda - 1) = (1 - \lambda)$, $v = w$ and $w = a$ in $\lambda v + (1 - \lambda)w \in A$, then $(1 - \lambda)w + (1 - (1 - \lambda))a = (-1)(\lambda - 1)w + \lambda a = \lambda a - (\lambda - 1)w \in A$, therefore $\lambda a - (\lambda - 1)w - w = \lambda a - \lambda w \in (-w) + A$.

Therefore $(-w) + A$ is a subapce of V and $w + (-w) + A$ is a translate.

□

Exercise 3.10. Let $A = a + U$ and $B = b + W$ where $a, b \in V$, $U, W \subseteq V$ are subspaces. Show that $A \cap B$ is either a translate of some subspace of V or an empty space.

Proof. Suppose $A \cap B \neq \emptyset$, we claim that $A \cap B$ is a translate of $U \cap W$, more specifcly, for any $a + u_0 = b + w_0 \in A \cap B$, we claim that $A \cap B = (a + u_0) + U \cap W$.

For any $u = w \in U \cap W$, we have $(a + u_0) + u = a + (u_0 + u) \in a + U$, similarly, we have $(b + w_0) + w = b + (w_0 + w) \in b + W$, therefore $(a + u_0) + (U \cap W) \subseteq A \cap B$.

For any $a + u = b + w \in A \cap B$, we have $a + u - (a + u_0) = u - u_0 \in U$ and $b + w - (b + w_0) = w - w_0 \in W$, therefore $A \cap B \subseteq (a + u_0) + (U \cap W)$. □

Exercise 3.12. Let $v_0, \dots, v_{m-1} \in V$ and

$$A = \{ \lambda_0 v_0 + \dots + \lambda_{m-1} v_{m-1} \mid \lambda_i \in F \text{ and } \lambda_0 + \dots + \lambda_{m-1} = 1 \}$$

1. Show that A is a translate of a subspace of V .
2. If B a translate of a subspace of V such that $v_0, \dots, v_{m-1} \in B$, show that $A \subseteq B$.
3. Base on (1), show that the dimension of such subspace is less then m .

Proof.

- If A is a translate of a subspace of V , say B , then for any $a \in A$, we have $A = a + B$. Therefore $B = (-a) + A$, we may pick $a = v_0$, we find that for any $b \in B$, it is in form $(-1)v_0 + \lambda_0 v_0 + \dots + \lambda_{m-1} v_{m-1}$ where $\lambda_0 + \dots + \lambda_{m-1} = 1$, which implies $(-1) + \lambda_0 + \dots + \lambda_{m-1} = 0$. Then we claim $B = \{ \lambda_0 v_0 + \dots + \lambda_{m-1} v_{m-1} \mid \lambda_0 + \dots + \lambda_{m-1} = 0 \}$ is a subspace and $A = v_0 + B$.

•

□

Exercise 3.16. Let $\varphi \in \mathcal{L}(V, F)$ where $\varphi \neq 0$, show that $\dim(V/(\text{null } \varphi)) = 1$.

Proof. For any non-zero $v + \text{null } \varphi, w + \text{null } \varphi \in V/(\text{null } \varphi)$ (existence is guaranteed since $\varphi \neq 0$), since $\varphi(w) \in F$, then there is some λ such that $\lambda\varphi(w) = \varphi(v)$ cause $\varphi(v)$ and $\varphi(w)$ are non-zero, then $\varphi(\lambda w) = \varphi(v)$, which means $v + \text{null } T = (\lambda w) + \text{null } T$, therefore $\dim(V/\text{null } \varphi)$ cause any two (non-zero) vectors are linear dependent. □

Exercise 3.17. Let $U \subseteq V$ a subspace such that $\dim(V/U) = 1$. Show that there is $\varphi \in \mathcal{L}(V, F)$ such that $\text{null } \varphi = U$.

Proof. We know there is an isomorphism $i \in \mathcal{L}(V/U, F)$ since $\dim(V/U) = \dim F = 1$, then $\varphi = i \circ \pi$ where $\pi \in \mathcal{L}(V, V/U)$. Since i is injective, $\text{null } \varphi = \text{null } \pi = U$. □

Exercise 3.18. Explain why a linear functional is either surjective or 0.

Proof. Cause $\dim F = 1$. □

Exercise 3.6. Let $\varphi, \beta \in V'$, show that $\text{null } \varphi \subseteq \text{null } \beta \iff \exists c \in F, \beta = c\varphi$.

Proof.

- ($\Rightarrow$) For any $v \notin \text{null } \beta$, we have $\beta(v) = \beta(v)(\varphi(v))^{-1}\varphi(v)$ we claim that $\beta = \beta(v)(\varphi(v))^{-1}\varphi$. We may denote $\beta(v)(\varphi(v))^{-1}$ by c . For any $v, w \notin \text{null } \beta$, we have $\beta(v) = a\varphi(v)$ and $\beta(w) = b\varphi(w)$, we want to show that $a = b$, which can be proven by:

$$\begin{aligned} a &= b \\ \frac{\beta(v)}{\varphi(v)} &= \frac{\beta(w)}{\varphi(w)} \\ \beta(v)\varphi(w) &= \beta(w)\varphi(v) \\ \beta(\varphi(w)v) &= \beta(\varphi(v)w) \end{aligned}$$

which is equivalent to $\varphi(w)v - \varphi(v)w \in \text{null } \beta$, then:

$$\begin{aligned} &\varphi(\varphi(w)v - \varphi(v)w) \\ &= \varphi(\varphi(w)v) - \varphi(\varphi(v)w) \\ &= \varphi(w)\varphi(v) - \varphi(v)\varphi(w) \\ &= 0 \end{aligned}$$

therefore $\varphi(w)v - \varphi(v)w \in \text{null } \varphi \subseteq \text{null } \beta$, thus $a = b$.

The case $v \in \text{null } \beta$ is trivial.

- ($\Leftarrow$) For any $v \in \text{null } \varphi$, $\beta(v) = c\varphi(v) = 0$, therefore $v \in \text{null } \beta$, thus $\text{null } \varphi \subseteq \text{null } \beta$.

□

Exercise 3.7. Let $V_0, \dots, V_{m-1}$ are vector spaces, show that $V'_0 \times \dots \times V'_{m-1}$ and $(V_0 \times \dots \times V_{m-1})'$ are isomorphic.

Proof. Define $\psi(\varphi) = v_0 \mapsto \varphi(v_0, 0, \dots), \dots, v_{m-1} \mapsto \varphi(\dots, 0, v_{m-1})$ and $\psi^{-1}(\varphi_0, \dots, \varphi_{m-1}) = (v_0, \dots, v_{m-1}) \mapsto \varphi_0(v_0) + \dots + \varphi_{m-1}(v_{m-1})$.

For any $\alpha, \beta \in (V_0 \times \dots \times V_{m-1})'$ and $\lambda \in F$, we have

$$\begin{aligned} & \psi(\alpha + \beta)_i \\ &= v_i \mapsto (\alpha + \beta)(\dots, v_i, \dots) \\ &= v_i \mapsto \alpha(\dots, v_i, \dots) + \beta(\dots, v_i, \dots) \\ &= (v_i \mapsto \alpha(\dots, v_i, \dots)) + (v_i \mapsto \beta(\dots, v_i, \dots)) \\ &= \psi(\alpha)_i + \psi(\beta)_i \end{aligned}$$

and $(\lambda\psi(\alpha))_i = \lambda(v_i \mapsto \alpha(v_i)) = v_i \mapsto \lambda\alpha(v_i) = \psi(\lambda\alpha)_i$ Therefore ψ is a linear map.

For any $\alpha, \beta \in V'_0 \times \dots \times V'_{m-1}$ and $\lambda \in F$, we have:

$$\begin{aligned} & \psi^{-1}(\alpha + \beta) \\ &= (v_0, \dots, v_{m-1}) \mapsto (\alpha + \beta)(v_0) + \dots + (\alpha + \beta)(v_{m-1}) \\ &= (v_0, \dots, v_{m-1}) \mapsto \alpha(v_0) + \beta(v_0) + \dots \\ &= ((v_0, \dots, v_{m-1}) \mapsto \alpha(v_0) + \dots) + ((v_0, \dots, v_{m-1}) \mapsto \beta(v_0) + \dots) \\ &= \psi^{-1}(\alpha) + \psi^{-1}(\beta) \end{aligned}$$

and

$$\begin{aligned} & \lambda\psi^{-1}(\alpha) \\ &= \lambda((v_0, \dots, v_{m-1}) \mapsto \alpha(v_0) + \dots) \\ &= (v_0, \dots, v_{m-1}) \mapsto \lambda(\alpha(v_0) + \dots) \\ &= (v_0, \dots, v_{m-1}) \mapsto (\lambda\alpha(v_0)) + \dots \\ &= \psi^{-1}(\lambda\alpha) \end{aligned}$$

thus ψ^{-1} is a linear map.

We will show that ψ^{-1} is the inverse of ψ then ψ is an isomorphism. For any $\varphi \in (V_0 \times \cdots \times V_{m-1})'$,

$$\begin{aligned}
& \psi^{-1}(\psi(\varphi)) \\
&= v_0, \dots, v_{m-1} \mapsto \psi(\varphi)_0(v_0) + \cdots \\
&= v_0, \dots, v_{m-1} \mapsto \varphi(v_0, 0, \dots) + \cdots + \varphi(\dots, 0, v_{m-1}) \\
&= v_0, \dots, v_{m-1} \mapsto \varphi(v_0, \dots, v_{m-1}) \\
&= \varphi
\end{aligned}$$

and for any $\varphi \in V'_0 \times \cdots \times V'_{m-1}$,

$$\begin{aligned}
& \psi(\psi^{-1}(\varphi)) \\
&= v_0 \mapsto \psi^{-1}(\varphi)(v_0, 0, \dots), \dots \\
&= v_0 \mapsto \varphi_0(v_0), \dots \\
&= \varphi_0, \dots, \varphi_{m-1} \\
&= \varphi
\end{aligned}$$

□

Exercise 3.16. Let W a finite vector space, $T \in \mathcal{L}(V, W)$, show that

$$T' = 0 \iff T = 0$$

Proof.

- ($\Rightarrow$) Suppose $T \neq 0$, then we can always find $\varphi \in \mathcal{L}(W, F)$ which $\varphi(\text{range } T) \neq 0$, then $\varphi \circ T \neq 0$.
- ($\Leftarrow$) Trivial.

□

Exercise 3.17. Let V, W are finite vector spaces, $T \in \mathcal{L}(V, W)$. Show that T is invertible $\iff T'$ is invertible.

Proof. Since T is invertible, then T is injective, therefore T' is surjective. Similarly, T' is injective since T is surjective. Therefore T' is invertible. □

Exercise 3.18. Let V, W are finite vector spaces, show that the mapping $\varphi(T) = T'$ is an isomorphism between $\mathcal{L}(V, W)$ and $\mathcal{L}(W', V')$.

Proof. Since V and W are finite, we only need to show that φ is injective or surjective. We will show that φ is injective.

For any $\varphi(T) = T' \in \mathcal{L}(W', V')$, we know $T = 0 \iff T' = 0$, therefore $\text{null } \varphi = \{0\}$, thus φ is injective.

I was wonder if I can prove this by $\varphi(S)(\text{id}) = \varphi(T)(\text{id}) \implies S = T$. This may work if the codomain restriction doesn't lose information, i.e. only restrict to a superset of its range, therefore the restriction is one-to-one. $\square$

Exercise 3.21. Let V finite and $U, W \subseteq V$ are subspaces.

1. Show that $W^0 \subseteq U^0 \iff U \subseteq W$
2. Show that $W^0 = U^0 \iff U = W$

Proof. The second statement can be easy proved by the first one.

- ($\Rightarrow$) We can always find a $f \in \mathcal{L}(W, F)$ such that $\text{null } f = W$, then $f(U) = \{0\}$ since $f \in W^0 \subseteq U^0$, therefore $U \subseteq \text{null } f = W$.
- ($\Leftarrow$) For any $\varphi \in W^0$, we know $W \subseteq \text{null } \varphi$, then $U \subseteq W \subseteq \text{null } \varphi$, therefore $\varphi \in U^0$, thus $W^0 \subseteq U^0$.

$\square$

Exercise 3.22. Let V finite and $U, W \subseteq V$ are subspaces. Show that:

- $(U + W)^0 = U^0 \cap W^0$
- $(U \cap W)^0 = U^0 + W^0$

Proof.

- For any $\varphi \in (U + W)^0$ we have $U + W \subseteq \text{null } \varphi$, then $U \subseteq U + W \subseteq \text{null } \varphi$ and $W \subseteq U + W$, therefore $\varphi \in U^0 \cap W^0$.

For any $\varphi \in U^0 \cap W^0$, we have $U \subseteq \text{null } \varphi$ and $W \subseteq \text{null } \varphi$. For any $u + w \in U + W$, we have $\varphi(u + w) = \varphi(u) + \varphi(w) = 0 + 0 = 0$, therefore $U + W \subseteq \text{null } \varphi$, thus $\varphi \in (U + W)^0$.

- For any $su + tw \in U^0 + W^0$, for any $v \in U \cap W$, we have $su(v) + tw(v) = s0 + t0$ since $v \in U$ and $v \in W$. Therefore we have an injective map

(also linear, this map just produce what it receive) from $U^0 + W^0$ to $(U \cap W)^0$. We have:

$$\begin{aligned}
& \dim(U^0 + W^0) \\
&= \dim U^0 + \dim W^0 - \dim(U^0 \cap W^0) \\
&= \dim V - \dim U + \dim V - \dim W - \dim(U + W)^0 \\
&= \dim V - \dim U + \dim V - \dim W - (\dim V - \dim(U + W)) \\
&= \dim V - \dim U - \dim W + (\dim U + \dim W - \dim(U \cap W)) \\
&= \dim V - \dim(U \cap W) \\
&= \dim(U \cap W)^0
\end{aligned}$$

therefore $(U \cap W)^0 = U^0 + W^0$.

□

Exercise 3.23. Let V finite and $\varphi_0, \dots, \varphi_{m-1} \in V'$. Show that the following sets are equal to each others:

- $\text{span}(\varphi_0, \dots, \varphi_{m-1})$
- $((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}))^0$
- $\{ \varphi \in V' \mid (\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}) \subseteq \text{null } \varphi \}$

Proof.

- $((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}))^0 = (\text{null } \varphi_0)^0 + \dots + (\text{null } \varphi_{m-1})^0$, then $\text{span}(\varphi_i) \subseteq (\text{null } \varphi_i)^0$ therefore $\text{span}(\varphi_0, \dots, \varphi_{m-1}) \subseteq ((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}))^0$.

For any $\varphi \in \text{span}(\varphi_0, \dots, \varphi_{m-1})$, we have $\varphi(v) = \varphi_0(v) + \dots + \varphi_{m-1}(v) = 0 + \dots + 0 = 0$ for any $v \in (\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1})$, therefore $\varphi \in ((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}))^0$.

- Last two sets are definitional equal.

□

Exercise 3.24. Let V finite and $v_0, \dots, v_{m-1} \in V$.

Define $\Gamma(\varphi) = (\varphi(v_0), \dots, \varphi(v_{m-1})) : V' \rightarrow F^m$, show that:

- $v_0, \dots, v_{m-1}$ spans $V \iff \Gamma$ is injective.

- $v_0, \dots, v_{m-1}$ is linear independent $\iff \Gamma$ is surjective.

Proof.

- $(\Rightarrow)$ Suppose $\Gamma(\alpha) = \Gamma(\beta)$, then for all $v \in V$ can be factorized into $\lambda_0 v_0 + \dots + \lambda_{m-1} v_{m-1}$, then $\alpha(v) = \alpha(\lambda_0 v_0 + \dots + \lambda_{m-1} v_{m-1}) = \beta(\lambda_0 v_0 + \dots + \lambda_{m-1} v_{m-1}) = \beta(v)$ since $\Gamma(\alpha) = \Gamma(\beta)$ and α and β are linear map, thus $\alpha = \beta$.
 $(\Rightarrow)$ We first make $v_0, \dots, v_{m-1}$ linear independent, say $v_0, \dots, v_{k-1}$, then for any $w \in V$ such that $v_0, \dots, v_{k-1}, w$ is linear independent, then we have its dual basis $\varphi_0, \dots, \varphi_{k-1}, \psi$. Consider $\Gamma(\psi)$, by definition, we know $\Gamma(\psi) = (\psi(v_0), \dots) = (0, \dots)$ then $\psi = 0$ since Γ is injective, which contradicts our assumption. Therefore $v_0, \dots, v_{k-1}$ spans V .
- $(\Rightarrow)$ Consider the dual basis of $v_0, \dots, v_{m-1}$, then Γ is surjective since we have the standard basis of F^m .
 $(\Leftarrow)$ Γ is surjective implies we have $\varphi_0, \dots, \varphi_{m-1}$ such that $\Gamma(\varphi_i) = (\dots, 1, \dots)$, which means $v_0, \dots, v_{m-1}$ is linear independent.

□

Exercise 3.25. Let V finite and $\varphi_0, \dots, \varphi_{m-1} \in V'$.

Define $\Gamma(v) = (\varphi_0(v), \dots, \varphi_{m-1}(v)) : V \rightarrow F^m$. Show that

- $\varphi_0, \dots, \varphi_{m-1}$ spans V' $\iff \Gamma$ is injective
- $\varphi_0, \dots, \varphi_{m-1}$ is linear independent $\iff \Gamma$ is surjective

Proof.

- $(\Rightarrow)$ Suppose $\Gamma(v) = \Gamma(w)$, then $\varphi_i(v) = \varphi_i(w)$, which means $\varphi_i(v - w) = 0$ for all i . If $v - w \neq 0$, then $((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}))^0 \neq \{0\}$, thus $\varphi_0, \dots, \varphi_{m-1}$ doesn't span V' .
 $(\Leftarrow)$ $(\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}) = \{0\}$ since Γ is injective. therefore $\text{span}(\varphi_0, \dots, \varphi_{m-1}) = ((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}))^0 = (\{0\})^0 = V'$
- $(\Rightarrow)$ We may treat Γ as the following matrix:

$$\begin{bmatrix} \varphi_0 \\ \vdots \\ \varphi_{m-1} \end{bmatrix}$$

which line rank is m since $\varphi_0, \dots, \varphi_{m-1}$ is linear independent, therefore its column rank is m , thus $\dim \text{range } \Gamma = m = \dim F^m$, then Γ is surjective.

($\Leftarrow$) It seems the proof of ($\Rightarrow$) also works here.

□

Exercise 3.26. Let V finite, and $\Omega \subseteq V'$ a subspace. Show that

$$\Omega = \{ v \in V \mid \varphi(v) = 0 \quad \forall \varphi \in \Omega \}^0$$

Proof. This construction looks like an inverse of $-^0$.

We may rewrite the equation to $\Omega = (\bigcap_{\varphi \in \Omega} \text{null } \varphi)^0$, then $\Omega = \text{span}(\varphi) \forall \varphi \in \Omega$, which is trivial. □

Exercise 3.28. Let V finite and $\varphi_0, \dots, \varphi_{m-1}$ is linear independent. Show that

$$\dim((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1})) = \dim V - m$$

Proof.

$$\begin{aligned} m &= \dim \text{span}(\varphi_0, \dots, \varphi_{m-1}) \\ &= \dim((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1}))^0 \\ &= \dim V - \dim((\text{null } \varphi_0) \cap \dots \cap (\text{null } \varphi_{m-1})) \end{aligned}$$

□

Exercise 3.30. Let V finite and $\varphi_0, \dots, \varphi_{m-1}$ a basis of V' . Show that there is a basis of V which dual basis is $\varphi_0, \dots, \varphi_{m-1}$.

Proof. Since $\varphi_0, \dots, \varphi_{m-1}$ spans V' and linear independent, we know Γ is both injective and surjective. Consider $v_0, \dots, v_{m-1}$ such that $\Gamma(v_i) = (\dots, 0, 1, 0, \dots)$. We claim $v_0, \dots, v_{m-1}$ is a basis of V and which dual basis is $\varphi_0, \dots, \varphi_{m-1}$.

The second part is trivial by the way construct them. For the first part, $v_0, \dots, v_{m-1}$ is linear independent since $(\dots, 0, 1, 0, \dots)$ is linear independent, and $v_0, \dots, v_{m-1}$ spans V since $\dim V = \dim V' = m$. □

Exercise 3.31. Let $U \subseteq V$ a subspace and $i(u) = u : U \rightarrow V$. Then $i' \in \mathcal{L}(V', U')$, show that:

1. $\text{null } i' = U^0$
2. $\text{range } i' = U'$ if V is finite
3. $\tilde{i}'$ is an isomorphism between V'/U^0 and U' if V is finite

Proof.

- For any $\varphi \in \text{null } i'$, $\varphi \circ i = 0$, therefore $\text{range } i = U \subseteq \text{null } \varphi$, thus $\varphi \in U^0$.

For any $\varphi \in U^0$, $\varphi \circ i = 0$ since $\text{range } i = U \subseteq \text{null } \varphi$.

- Suppose V is finite, then i' is surjective since i' is injective, therefore $\text{range } i' = U'$.
- $\tilde{i}'(\varphi + U^0) = i'(\varphi)$ is surjective since i' is surjective. Then $\dim(V'/U^0) = \dim V' - \dim U^0 = \dim V - (\dim V - \dim U) = \dim U = \dim U'$, therefore $\tilde{i}'$ is an isomorphism.

□

Exercise 3.32. We denote V'' as the **double dual space of V** , defined by $V'' = (V')'$. Define $\Lambda(v)(\varphi) = \varphi(v) : V \rightarrow V''$

Show that:

1. $\Lambda \in \mathcal{L}(V, V'')$
2. Let $T \in \mathcal{L}(V)$, then $T'' \circ \Lambda = \Lambda \circ T$ where $T'' = (T')'$.
3. Λ is an isomorphism if V is finite.

Proof.

- For any $v, w \in V$ and $\lambda \in F$, we have $(\Lambda(v) + \Lambda(w))(\varphi) = \Lambda(v)(\varphi) + \Lambda(w)(\varphi) = \varphi(v) + \varphi(w) = \varphi(v + w) = \Lambda(v + w)(\varphi)$ and $(\lambda\Lambda(v))(\varphi) = \lambda(\Lambda(v)(\varphi)) = \lambda(\varphi(v)) = \varphi(\lambda v) = \Lambda(\lambda v)(\varphi)$.

- For any $v \in V$,

$$\begin{aligned}
& (T'' \circ \Lambda)(v)(\varphi) \\
&= (T''(\Lambda(v)))(\varphi) \\
&= ((\Lambda(v)) \circ T')(\varphi) \\
&= \Lambda(v)(T'(\varphi)) \\
&= \Lambda(v)(\varphi \circ T) \\
&= (\varphi \circ T)(v) \\
&= \varphi(T(v)) \\
&= \Lambda(T(v))(\varphi) \\
&= (\Lambda \circ T)(v)(\varphi)
\end{aligned}$$

- Suppose $\Lambda(v) = \Lambda(w)$, that is, $\Lambda(v)(\varphi) = \varphi(v) = \varphi(w) = \Lambda(w)(\varphi)$ for all $\varphi \in V'$. Let $\varphi_0, \dots, \varphi_{m-1}$ the dual basis of some basis of V , then $v = \varphi_0(v)v_0 + \dots + \varphi_{m-1}(v)v_{m-1} = \varphi_0(w)v_0 + \dots + \varphi_{m-1}(w)v_{m-1} = w$. Therefore Λ is injective, thus surjective and isomorphism since $\dim V = \dim V''$.

□

Exercise 3.33. Let $U \subseteq V$ a subspace and $\pi : V \rightarrow V/U$ the quotient map, then $\pi' \in \mathcal{L}((V/U)', V')$.

1. Show that π' is injective.
2. Show that $\text{range } \pi' = U^0$.
3. Conclude that π' is an isomorphism between $(V/U)'$ and U^0 .

Proof.

- π is surjective, therefore π' is injective. The statement is true even V or V/U may be infinite, cause the proof about surjective-implies-epimorphism doesn't require that the codomain is finite but epimorphism-implies-surjective does.

We may prove those theorem again, but with weaker assumption. For any $\pi'(\varphi) = \pi'(\psi)$, we have $\varphi \circ \pi = \psi \circ \pi$. For any $v + U \in V/U$, there is $v \in V$ such that $\pi(v) = v + U$ since π is surjective. Therefore $\varphi(\pi(v)) = \psi(\pi(v))$ for all $\pi(v) = v + U \in V/U$, thus $\varphi = \psi$.

Therefore π' is injective.

- $\text{range } \pi' = (\text{null } \pi)^0 = U^0$.
- Trivial.

□

Exercise 3.7. Let m a non-negative integer and $z_0, \dots, z_m \in F$ are different to each others, $w_0, \dots, w_m \in F$, show that there is a unique $p \in \mathcal{P}_m(F)$ such that $p(z_k) = w_k$ holds for all $0 \leq k \leq m$.

Proof. Define $\Gamma(p) = (p(z_0), \dots, p(z_m)) : \mathcal{P}_m(F) \rightarrow F^{m+1}$, we will show that Γ is injective, therefore an isomorphism.

Suppose $\Gamma(p) = \Gamma(q)$, then $p(z_k) = q(z_k)$ for all k , therefore $(p-q)(z_k) = 0$ for all k . This means $p - q$ has $m + 1$ zeros but $\deg(p - q) \leq m$, therefore $p - q = 0$ and $p = q$.

Then there is a unique $p \in \mathcal{P}_m(F)$ such that $\Gamma(p) = (w_0, \dots, w_m)$. □

Exercise 3.9. Let $P \in \mathcal{L}(V)$, such that $P^2 = P$. Suppose λ an eigenvalue of P , show that $\lambda = 0$ or $\lambda = 1$.

Proof. Suppose $P(v) = \lambda v$ for some non-zero $v \in V$, then $P(v) = PP(v) = P(\lambda v)$, therefore $P((\lambda - 1)v) = 0$. thus $(\lambda - 1)v \in \text{null } P$. We may suppose $\lambda \neq 1$, then $(\frac{1}{\lambda-1})(\lambda - 1)v = v \in \text{null } P$, therefore $P(v) = 0$, thus $\lambda = 0$ cause $v \neq 0$. □

Exercise 3.10. Let $T(p) = p' : \mathcal{P}(\mathbb{R}) \rightarrow \mathcal{P}(\mathbb{R})$. Find all eigenvalues and eigenvectors of T .

Proof. Suppose $T(p) = p' = \lambda p$, then $\deg p = 0$, otherwise the degree doesn't match. For any $p \in \mathcal{P}(\mathbb{R})$ such that $\deg p = 0$, we have $p' = 0 = 0p$. □

Exercise 3.12. Let $V = U \oplus W$ where U and W are non-zero subspaces. Define $P(u + w) = u$ for all $u \in U$ and $w \in W$. Find all eigenvalue and eigenvector of P .

Proof. We can see $P^2 = P$, since for any $u + w \in V$, we have $P(P(u + w)) = P(u) = u = P(u + w)$, therefore $\lambda = 0$ and $\lambda = 1$ are eigenvalues of P , $P(u) = 1u$ and $P(w) = 0w$ are eigenvectors of P . □

Exercise 3.13. Let $T \in \mathcal{L}(V)$ and $S \in \mathcal{L}(V)$, where S is invertible.

- Show that T has the same eigenvalue of $S^{-1}TS$.

- What is the relationship between the eigenvector of T and the eigenvector of $S^{-1}TS$.

Proof.

- For any $T(v) = \lambda v$ where $v \in V$ and $\lambda \in F$, let $S(w) = v$, then $S^{-1}TS(w) = S^{-1}(T(Sw)) = S^{-1}(\lambda v) = \lambda S^{-1}(v) = \lambda w$, thus λ is an eigenvalue of $S^{-1}TS$.
- $S(w) = v$ where v is an eigenvector of T and w is the corresponding eigenvector of $S^{-1}TS$.

□

Exercise 3.15. Let V finite, $T \in \mathcal{L}(V)$, $\lambda \in F$. Show that λ is an eigenvalue of $T \iff \lambda$ is an eigenvalue of T' .

Proof.

- ($\Rightarrow$) Suppose $Tv = \lambda v$, we will show $T' - \lambda I$ is not surjective (Note that $I \in \mathcal{L}(V')$).

For any $\varphi \in V'$, we have:

$$\begin{aligned} & (T' - \lambda I)(\varphi) \\ &= T'(\varphi) - \lambda\varphi \\ &= \varphi \circ T - \lambda\varphi \end{aligned}$$

then

$$\begin{aligned} & (\varphi \circ T - \lambda\varphi)(v) \\ &= (\varphi \circ T)(v) - (\lambda\varphi)(v) \\ &= \varphi(Tv) - \lambda(\varphi(v)) \\ &= \varphi(\lambda v) - \lambda(\varphi(v)) \\ &= 0 \end{aligned}$$

This means $\text{range } T' \neq V'$ cause any $\psi \in V'$ where $\psi(v) \neq 0$ is not in $\text{range } T'$. Therefore λ is an eigenvalue of T' .

□

Exercise 3.22. Let $T \in \mathcal{L}(V)$ and non-zero $v, w \in V$ such that

$$Tu = 3w \quad \text{and} \quad Tw = 3u$$

Show that 3 or -3 is the eigenvalue of T .

Proof. Since v and w are non-zero, then one of $u + w$ and $u - w$ is non-zero.

We have $T(u + w) = 3w + 3u = 3(u + w)$ and $T(u - w) = 3w - (3u) = (-3)(u - w)$. $\square$

Exercise 3.23. Let V finite, and $S, T \in \mathcal{L}(V)$, show that ST and TS have the same eigenvalues.

Proof. For any $ST(v) = \lambda v$ where $v \neq 0$, we have $TST(v) = T(\lambda v)$ then $TS(Tv) = \lambda(Tv)$.

- If $Tv = 0$, then $STv = 0 = \lambda v$, thus $\lambda = 0$ since $v \neq 0$. then λ is an eigenvalue of TS since $\text{null } TS \neq 0$ ($\text{null } T \neq 0$, and S is an operator of V , therefore, S is injective or not doesn't affect our conclusion).

If $Tv \neq 0$, then $TS(Tv) = \lambda(Tv)$.

- Ditto.

$\square$

Exercise 3.26. Let $T \in \mathcal{L}(V)$ and any non-zero $v \in V$ we have $Tv = cv$ for some c . Show that $T = \lambda I$.

Proof. Let non-zero $v, w \in V$, we have $Tv = sv$ and $Tw = tw$, then $T(v + w) = \lambda(v + w) = \lambda v + \lambda w = sv + tw = T(v) + T(w)$. Then $\lambda v + \lambda w - tw = sv$.

- If $w \in \text{span}(v)$, then $w = cv$, therefore $Tw = T(cv) = tcv = cTv = csv$, thus $t = s$.
- If $w \notin \text{span}(v)$, then $\lambda = t$ (otherwise $\lambda v + (\lambda - t)w = sv$), therefore $\lambda v = sv$ and $\lambda = s$, thus $s = t$.

$\square$

Exercise 3.27. Let V finite and $1 \leq k \leq \dim V - 1$. Let $T \in \mathcal{L}(V)$ such that any subspace of V with k dimension is invariant under T . Show that $T = \lambda I$ for some λ .

Proof. For any $v \in V$, we have $Tv = w$ where $w \in \text{span}(v)$.

- If $w = 0$, then $Tv = 0v$.
- If $w \neq 0$, then $w = \lambda v$ since $w \in \text{span}(v)$, then $Tv = \lambda v$.

Thus $T = \lambda I$ by the previous exercise. $\square$

Exercise 3.30. Let $T \in \mathcal{L}(V)$ and $(T - 2I)(T - 3I)(T - 4I) = 0$. Show that 2 or 3 or 4 is the eigenvalue of T .

Proof. Suppose 2 is not an eigenvalue of T , then $(T - 2I)$ is injective, thus $(T - 3I)(T - 4I)$ must map all $v \in V$ to 0. Similarly, we can show that $(T - 4I) = 0$ if 3 is not an eigenvalue of T . $\square$

Exercise 3.31. Find $T \in \mathcal{L}(\mathcal{R}^2)$ such that $T^4 = -I$.

Proof. We may treat $-I$ as rotating the vector 180 degrees, then T rotates a vector 45 degrees, which matrix is:

$$\mathcal{M}(T) = \frac{\sqrt{2}}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

$\square$

Exercise 3.32. Let $T \in \mathcal{L}(V)$ with no eigenvalue and $T^4 = I$. Show that $T^2 = -I$.

Proof. We will show that $T^2(v) = -v$ for any $v \in V$. Consider $T^2(T^2(v) + v) = v + T^2(v)$, we will show that $T^2(v) + v = 0$. Suppose $w \in V$ and $T^2(w) = w$, we may let $T(w) = u$, then $T^2(w) = Tu = w$. Consider $T(w + u) = T(w) + T(u) = u + w$, then $w + u = 0$ since T has no eigenvalue, therefore $u = -w$ and $T(w) = -w$. Again $w = 0$ since T has no eigenvalue. Therefore $T^2(w) = w$ implies $w = 0$, thus $T^2(v) + v = 0$ and $T^2(v) = -v$ for any $v \in V$. $\square$

Exercise 3.33. Let $T \in \mathcal{L}(V)$ and $m \in \mathbb{N}^+$.

1. Show that T is injective $\iff T^m$ is injective
2. Show that T is surjective $\iff T^m$ is surjective.

Proof. Recall that $T^0 = I$.

- ($\Rightarrow$) For any $T^m(v) = T^m(w)$ we have $T^{m-1}(v) = T^{m-1}(w)$ and so on, we will get $v = w$.
 ($\Leftarrow$) For any $T(v) = T(w)$, we have $T^{m-1}(T(v)) = T^{m-1}(T(w))$, then $T^m(v) = T^m(w)$ and $v = w$.
- ($\Rightarrow$) For any $w \in V$, we have $T(v) = w$, then we have $T(u) = v$ and now $T(T(u)) = w$, continue this progress until we get $T^m(r) = w$.
 ($\Leftarrow$) For any $w \in V$, we have $T^m(v) = w$, therefore $T(T^{m-1}(v)) = w$.

□

Exercise 3.34. Let V finite and $v_0, \dots, v_{m-1} \in V$. Show that $v_0, \dots, v_{m-1}$ is linear independent $\iff$ there is $T \in \mathcal{L}(V)$ such that $v_0, \dots, v_{m-1}$ are eigenvectors of distinct eigenvalues of T .

Proof.

- ($\Rightarrow$) Consider $v_0, \dots, v_{k-1}$ a basis of V , then $T(\lambda_0 v_0 + \dots + \lambda_{k-1} v_{k-1}) = 1\lambda_0 v_0 + 2\lambda_1 v_1 + \dots + m\lambda_{m-1} v_{m-1}$ where $T(v_i) = (i+1)v_i$.
- ($\Leftarrow$) Trivial, since eigenvectors of distinct eigenvalues are linear independent.

□

Exercise 3.37. Let V finite and $T \in \mathcal{L}(V)$. Define $\mathcal{A}(S) = TS : \mathcal{L}(V) \rightarrow \mathcal{L}(V)$. Show that T has the same eigenvalues as $\mathcal{A}$.

Proof.

- ($\subseteq$) For any eigenvalue λ of T , we can find $S \in \mathcal{L}(V)$ such that $\text{range } S = \{ v \in V \mid Tv = \lambda v \}$ (it is easy to show that such set is a subspace). Then for any $v \in V$, $(TS)v = T(Sv) = \lambda(Sv) = (\lambda S)v$ thus $\mathcal{A}(S) = TS = \lambda S$.
- ($\supseteq$) For any eigenvalue λ of $\mathcal{A}$, then we have $\mathcal{A}(S) = \lambda S$ for some non-zero $S \in \mathcal{L}(V)$. Then there is $v \in V$ such that $Sv \neq 0$, and $T(Sv) = (TS)v = (\lambda S)(v) = \lambda(Sv)$, thus λ is an eigenvalue of T .

□

Exercise 3.38. Let V finite and $T \in \mathcal{L}(V)$ and $U \subseteq V$ is invariant under T . A quotient operator $T/U \in \mathcal{L}(V/U)$ is defined by:

$$(T/U)(v + U) = Tv + U$$

for any $v \in V$.

1. Show that T/U is well-defined and T/U is an operator over V/U .
2. Show that each eigenvalue of T/U is also an eigenvalue of T .

Proof.

- Suppose $v + U = w + U$, then $(T/U)(v + U) = Tv + U$ and $(T/U)(w + U) = Tw + U$, we will show that $Tv - Tw \in U$. Note that $v + U = w + U$ implies $v - w \in U$, then $T(v - w) \in U$ since U is invariant under T , that is, for any $u \in U$, $Tu \in U$. Thus $Tv + U = Tw + U$.

Now we will show that T/U is a linear map, we can see:

$$\begin{aligned} & (T/U)(v + U) + (T/U)(w + U) \\ &= (Tv + U) + (Tw + U) \\ &= (Tv + Tw) + U \\ &= T(v + w) + U \\ &= (T/U)((v + w) + U) \end{aligned}$$

and

$$\begin{aligned} & \lambda(T/U)(v + U) \\ &= \lambda(Tv + U) \\ &= (\lambda(Tv)) + U \\ &= T(\lambda v) + U \\ &= (T/U)((\lambda v) + U) \\ &= (T/U)(\lambda(v + U)) \end{aligned}$$

- Suppose $(T/U)(v + U) = Tv + U = \lambda v + U$ where $v \notin U$, consider $T - \lambda I$, we will show that $T - \lambda I$ is not injective. We can see U is invariant under $T - \lambda I$, $Tu - \lambda u \in U$ cause U is invariant under T . We may suppose T is injective (thus surjective and invertible) on U (in other words, $T(U) = U$), otherwise the proof is complete. Then consider $(T - \lambda I)(v) = Tv - \lambda v \in U$ where $v \notin U$, thus $T - \lambda I$ is not injective.

□

Exercise 3.39. Let V finite and $T \in \mathcal{L}(V)$. Show that T has an eigenvalue $\iff$ there is a subspace of V with dimension $\dim V - 1$ which is invariant under T .

Proof.

- This part is hinted by AI. Suppose $Tv = \lambda v$, then consider $T - \lambda I$, we know $\text{range}(T - \lambda I)$ is invariant under T , since for any $Tw - \lambda w$, we have $T(Tw - \lambda w) = T(Tw) - T(\lambda w) = T(Tw) - \lambda(Tw)$. Then $\dim \text{range}(T - \lambda I) \leq \dim V - 1$ since $w \in \text{null } T - \lambda I$. Then consider $\text{null}(T - \lambda I) = \text{span}(v) \oplus W$, we have $\text{range}(T - \lambda I) \oplus W$ a subspace which is invariant under T .

The key is finding a smaller invariant subspace and expand it with null space, as any vector in null space always maps to 0, thus preserve the property of invariant.

- Suppose U is a subspace of V of dimension $\dim V - 1$ such that U is invariant under T , then $V = U \oplus \text{span}(v)$ for some $v \notin U$. We may suppose T is injective on U , otherwise the proof is complete ($\text{null } T \neq 0$). Consider $T(v)$, there are three cases:
 - $T(v) = \lambda v + 0u$, then the proof is complete.
 - $T(v) = 0v + u$, then T is not injective since there is $Tw = u$ where $w \in U$.
 - $T(v) = \lambda v + u$, then consider $T - \lambda I$. We have U is invariant under $T - \lambda I$ cause $Tu - \lambda u \in U$ by $Tu \in U$. Again, if $T - \lambda I$ is not injective on U , the proof is complete. Then $(T - \lambda I)v = Tv - \lambda v \in U = \lambda v + u - \lambda v = u \in U$, thus $T - \lambda I$ is not injective and λ is an eigenvalue of T .

□

Exercise 3.42. Let $T \in \mathcal{L}(F^n)$ defined by $T(x_1, x_2, \dots, x_n) = (x_1, 2x_2, \dots, nx_n)$.

1. Find all eigenvalues and eigenvectors of T .
2. Find all subspace of F^n which is invariant under T .

Proof.

- $1, 2, \dots, n$ and $(x_1, 0, \dots), (0, x_2, 0, \dots), \dots$
- We claim any subspace that is invariant under T is a direct sum of some spaces that spans by the standard basis, say $\text{span}(x_0) \oplus \dots \oplus \text{span}(x_k)$.

Let U a subspace that is invariant under T and $u \in U$, we have $T(u) = T(\lambda_1 x_1, \dots, \lambda_n x_n) = (\lambda_1 x_1, \dots, n \lambda_n x_n)$, then $T(u) - iu = ((1-i)(\lambda_1 x_1), (2-i)(\lambda_2 x_2), \dots, (n-1)(\lambda_i x_i)) \in U$ is a vector that is a linear combination of standard basis except x_i . Repeat this progress by apply $T - jI$ to $(T - iI)(u)$ with a different j , we can finally get a vector that is a scalar multiple of x_k . Thus $x_i \in U$ as long as there is $u \in U$ that the i th scalar of the linear combination of standard basis is not zero.

□

Exercise 3.43. Let $T \in \mathcal{L}(V)$. Show that 9 is an eigenvalue of $T^2 \iff 3$ or -3 is an eigenvalue of T .

Proof.

- $(\Rightarrow)$ We have $T^2 - 9I$ is not injective since 9 is an eigenvalue of T^2 , then $(T - 3I)(T + 3I) = T^2 - 9I$ is not injective means one of $T - 3I$ and $T + 3I$ is not injective, thus 3 or -3 is an eigenvalue of T .
- $(\Leftarrow)$ Similarly, we have $(T - 3I)(T + 3I)v = (T^2 - 9I)v = 0$ (if 3 is an eigenvalue of T) or $(T + 3I)(T - 3I)v = (T^2 - 9I)v = 0$ (if -3 is an eigenvalue of T).

□

Exercise 3.44. Let V a vector space over $\mathbb{C}$ and $T \in \mathcal{L}(V)$ has no eigenvalue. Show that any subspace of V that is invariant under T is either $\{0\}$ or infinite dimension.

Proof. Let $U \subseteq V$ a subspace that is invariant under T , and non-zero $u \in U$. We can repeatedly apply T to u , say $u, Tu, T^2u, \dots$. Suppose $k > 0$ is minimum such that $u, Tu, \dots, T^k u$ is linear dependent, we have $p \in \mathcal{P}(\mathbb{C})$ with $\deg p = k$ such that $p(T) = 0$. Clearly p is not constants, thus it has a zero since p is a polynomial of complex coefficient. Thus such zero is an eigenvalue of T .

□

Exercise 3.45. Let $n > 1$ an integer, and $T \in \mathcal{L}(F^n)$ is defined by:

$$T(x_0, \dots, x_{n-1}) = (x_0 + \dots + x_{n-1}, \dots, x_0 + \dots + x_{n-1})$$

- Find all eigenvalue and eigenvector of T .
- Find the minimal polynomial of T .

Proof.

- Observe that $\text{range } T = \text{span}((1, \dots, 1))$, thus $T(1, \dots, 1) = n(1, \dots, 1)$.
- Observe that $T(x_0, \dots, x_{n-1}) = (x_0 + \dots + x_{n-1})(1, \dots, 1)$ and $T^2(x_0, \dots, x_{n-1}) = n(x_0 + \dots + x_{n-1})(1, \dots, 1)$, thus $p(T) = nT - T^2 = 0$.

□

Exercise 4 is kinda hard, sorry.

Exercise 3.6. Let $T \in \mathcal{L}(F^2)$ is defined by $T(w, z) = (-z, w)$. Find the minimal polynomial of T .

Proof. Observe that $T^2(w, z) = T(-z, w) = (-w, -z) = (-1)(w, z)$, thus the minimal polynomial of T is $p(T) = I + T^2$. □

Exercise 3.7. • Given an example that the minimal polynomial of ST is not equal to TS 's.

- Suppose V is finite and $S, T \in \mathcal{L}(V)$. Show that the minimal polynomial of ST is equal to TS 's if one of S and T is invertible.

Hint: Show that S is invertible and $p \in \mathcal{P}(F)$ implies $p(TS) = S^{-1}p(ST)S$.

Proof.

- The idea is to find S, T such that $ST \neq 0$ but $TS = 0$. We can find $S(x, y) = (x, 0)$ and $T(x, y) = (y, 0)$ holds:

$$(ST)(x, y) = S(y, 0) = (y, 0)$$

$$(TS)(x, y) = T(x, 0) = (0, 0)$$

Thus the minimal polynomial of ST is not 0 but TS one does.

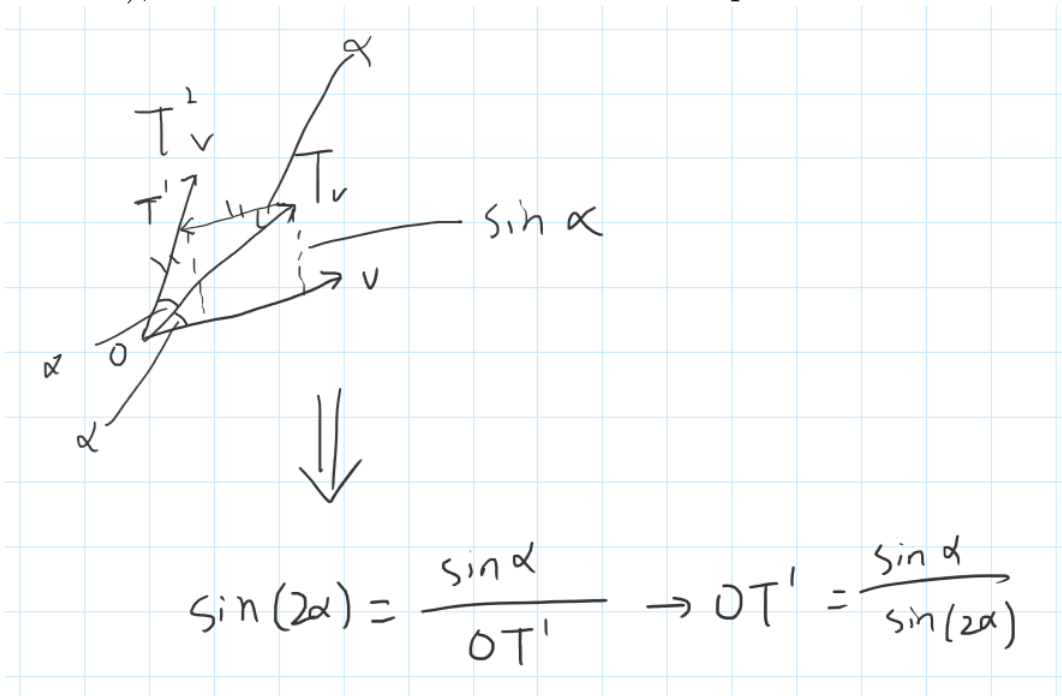
- Suppose S is invertible and $p \in \mathcal{L}(F)$ is the minimal polynomial of TS , then $p(TS) = S^{-1}p(ST)S$ since i -th term of $S^{-1}p(ST)S$ has form $S^{-1}c_i(ST)^iS = c_i(S^{-1}S)(TS)^{i-1}(TS) = c_i(TS)^i$. Thus $S^{-1}p(ST)S = 0$ and then $p(ST) = 0$. We will show that p is the minimal polynomial of ST , suppose $q \in \mathcal{L}(F)$ such that $q(ST) = 0$, then $0 = S^{-1}q(ST)S = q(TS)$, therefore $\deg q = \deg p$. Hence p is the minimal polynomial of ST .

□

Exercise 3.8. Let $T \in \mathcal{L}(R^2)$ is the operator that "rotates 1 degree counter-clockwise", find the minimal polynomial of T .

Note that it is **NOT** $x^{180} + 1$ even $T^{180} = -I$.

Proof. Note that there is some λ such that $Tv - \lambda v = \alpha T^2v$ (We can show that $\lambda = \alpha$), however the calculation is too complicate.



λ should be $\frac{\sin(1^\circ)}{\sin(2^\circ)}$, thus $p(T) = -\lambda I + T - \lambda T^2$.

We suppose all v below has length 1, thus $v = (\cos \theta, \sin \theta)$, this doesn't lose the generalizability since $p(T)(\alpha v) = \alpha(p(T)v)$.

For the first component of $p(T)v = -\lambda v + Tv - \lambda T^2v$, we have:

$$\begin{aligned}
& \frac{\sin(1^\circ)}{\sin(2^\circ)}(-\cos \theta - \cos(\theta + 2^\circ)) \\
&= \frac{\sin(1^\circ)}{\sin(2^\circ)}(-\cos \theta - (\cos \theta \cos(2^\circ) - \sin \theta \sin(2^\circ))) \\
&= \frac{\sin(1^\circ)}{\sin(2^\circ)}(-\cos \theta - \cos \theta \cos(2^\circ)) + \sin \theta \sin(1^\circ) \\
&= \cos \theta \frac{\sin(1^\circ)}{\sin(2^\circ)}(-1 - \cos(2^\circ)) + \sin \theta \sin(1^\circ)
\end{aligned}$$

where $\sin \theta \sin(1^\circ)$ cancels a part of $(Tv)_1 = \cos(\theta + 1^\circ) = \cos \theta \cos(1^\circ) - \sin \theta \sin(1^\circ)$. Thus we will show that $\frac{\sin(1^\circ)}{\sin(2^\circ)}(-1 - \cos(2^\circ)) = -\cos(1^\circ)$.

$$\begin{aligned}
& \frac{\sin(1^\circ)}{\sin(2^\circ)}(-1 - \cos(2^\circ)) \\
&= \frac{\sin(1^\circ)}{2 \sin(1^\circ) \cos(1^\circ)}(-(\cos^2(1^\circ) + \sin^2(1^\circ)) - \cos^2(1^\circ) + \sin^2(1^\circ)) \\
&= \frac{1}{2 \cos(1^\circ)}(-\cos^2(1^\circ) - \cos^2(1^\circ)) \\
&= \frac{1}{2 \cos(1^\circ)}(-2 \cos^2(1^\circ)) \\
&= -\cos(1^\circ)
\end{aligned}$$

For the second component of $p(T)v$, we have:

$$\begin{aligned}
& \frac{\sin(1^\circ)}{\sin(2^\circ)}(-\sin \theta - \sin(\theta + 2^\circ)) \\
&= \frac{\sin(1^\circ)}{\sin(2^\circ)}(-\sin \theta - \sin \theta \cos(2^\circ) - \cos \theta \sin(2^\circ)) \\
&= \sin \theta \frac{\sin(1^\circ)}{\sin(2^\circ)}(-1 - \cos(2^\circ)) - \cos \theta \sin(1^\circ)
\end{aligned}$$

similarly, we have $p(T)v_2 = \sin(\theta + 1^\circ) = \sin \theta \cos(1^\circ) + \cos \theta \sin(1^\circ)$ we will show that $\frac{\sin(1^\circ)}{\sin(2^\circ)}(-1 - \cos(2^\circ)) = -\cos(1^\circ)$, which is proven above. $\square$

Exercise 3.9. Let $T \in \mathcal{L}(V)$ such that for some basis of V , $\mathcal{M}(T)$ consists of rational numbers. Try to explain why the coefficients of the minimal polynomial of T is rational numbers.

Proof. I don't know, because $\mathbb{Q}$ is also a field? □

Exercise 3.11. Let V a vector space and $\dim V = 2$ and $T \in \mathcal{L}(V)$ such that $\mathcal{M}(T)$ for some basis of V is $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$. Show that:

- $T^2 - (a + d)T + (ad - bc)I = 0$
- the minimal polynomial of T is:

$$\begin{cases} z - a & \text{if } b = c = 0 \text{ and } a = d \\ z^2 - (a + d)z + (ad - bc) & \text{otherwise} \end{cases}$$

Proof.

•

$$\begin{aligned} & \mathcal{M}(T^2 - (a + d)T + (ad - bc)I) \\ &= \begin{bmatrix} a & c \\ b & d \end{bmatrix}^2 - (a + d) \begin{bmatrix} a & c \\ b & d \end{bmatrix} + (ad - bc)I \\ &= \begin{bmatrix} a^2 + bc & ac + bd \\ ab + bd & bc + d^2 \end{bmatrix} - \begin{bmatrix} a^2 + ad & ac + cd \\ ab + bd & ad + d^2 \end{bmatrix} + \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} \end{aligned}$$

- If $b = c = 0$ and $a = d$, then T is a scalar multiple of identity operator, thus $T = aI$ and $p(T) = -aI + T = 0$. Otherwise, $T^2 - (a + d)T + (ad - bc)I = 0$.

□

Exercise 3.13. Let V finite, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(F)$. Show that there is a unique $r \in \mathcal{P}(F)$ such that $p(T) = r(T)$ where $\deg p$ is less than the degree of the minimal polynomial of T .

Proof. Let q the minimal polynomial of T .

If $\deg p < \deg q$, then $r = p$. The uniqueness is guaranteed by $\deg p < \deg q$ (try $(p - s)(T)$ where $p(T) = s(T)$ and $\deg s < \deg q$).

If $\deg p \geq \deg q$, then $p = sq + r$ where $s, r \in \mathcal{P}(F)$ with $\deg r < \deg q$. Then $p(T) = s(T)q(T) + r(T) = r(T)$ since $q(T) = 0$. The uniqueness is guaranteed by the property of division. □

Exercise 3.14. Let V finite, $T \in \mathcal{L}(V)$ with minimal polynomial $p(z) = 4 + 5z - 6z^2 - 7z^3 + 2z^4 + z^5$. Find the minimal polynomial of T^{-1} .

Proof. Suppose p is the minimal polynomial of T , then we can repeatedly apply T^{-1} to $p(T)$, say $T^{-(\deg p)}(p(T))$, then it should be 0, and the coefficients are reversed, that is, $p_{\deg p}I + p_{\deg p-1}T^{-1} + \cdots + p_0(T^{-1})^{\deg p}$.

So the answer is $1 + 2z^1 - 7z^2 - 6z^3 + 5z^4 + 4z^5$. $\square$

Exercise 3.16. Let $a_0, \dots, a_{n-1} \in F$ and T an operator over F^n . Its matrix (about the standard basis) is:

$$\begin{bmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & 1 & \ddots & & -a_2 \\ & & \ddots & & \vdots \\ & & & 1 & 0 & -a_{n-2} \\ & & & & 1 & -a_{n-1} \end{bmatrix}$$

. Show that the minimal polynomial of T is:

$$p(z) = a_0 + a_1z + \cdots + a_{n-1}z^{n-1} + z^n$$

Proof. We first need some property of this matrix, we will see it moves all number to the left when we repeatedly self-multiply T . We can see the k -th column of T^p is equal to $k+1$ -th column of T^{p-1} , thus it is also equal to i -th column if T^j where $1 \leq i, j \leq n$ and $i+j = k+p$. In fact, j can be 0 and we have $T^0 = I$ and the property still holds.

Then, the i -th column of T^n is equal to n -th column (the last one) of T^i ,

and it is produced by $T^{i-1} \begin{bmatrix} -a_0 \\ -a_1 \\ \vdots \\ -a_{n-1} \end{bmatrix}$, which is equal to

$$T_i^n = -a_0T_1^{i-1} - a_1T_2^{i-1} - \cdots - a_{n-1}T_n^{i-1}$$

which is equal to $T^n v_i$ where v_i is i -th standard basis of F^n , that is, $\begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$. We

may rewrite the equation into

$$T^n v = -a_0 T^0 v - a_1 T^1 v - \cdots - a_{n-1} T^{n-1} v$$

where $T^0 v = T_i^0 = T_1^{i-1}$, $T^1 v = T_i^1 = T_2^{i-1}$ and so on.

Thus all v vector in standard basis has $p(T)v = 0$, thus $p(T) = 0$.

For minimal, we can see T is invertible, thus $p(T)v_1 = 0$ (recall that T moves number to the left, thus the first column of T^i is the i -th columns of T). means there is a (non-zero) linear combination of columns of T that is equal to 0. Thus $\deg p \geq n$ since T the columns are linear independent. $\square$

Exercise 3.17. Let V finite and $T \in \mathcal{L}(V)$ and p is the minimal polynomial of T . Let $\lambda \in F$, show that the minimal polynomial of $T - \lambda I$ is $q(z) = p(z + \lambda)$.

Proof. $q(T - \lambda I) = p((T - \lambda I) + \lambda I) = p(T) = 0$. Suppose r is the minimal polynomial of $T - \lambda I$, then $s(z) = r(z - \lambda)$ and $s(T) = r(T - \lambda I) = 0$, thus $\deg r = \deg p = \deg q$. $\square$

Exercise 3.18. Let V finite and $T \in \mathcal{L}(V)$, and p is the minimal polynomial of T . Let $\lambda \in F$ that $\lambda \neq 0$, show that the minimal polynomial of λT is $q(z) = \lambda^{\deg p} p(\frac{z}{\lambda})$.

Proof. $q(\lambda T) = \lambda^{\deg p} p(\frac{1}{\lambda}(\lambda T)) = \lambda^{\deg p} p(T) = 0$. $\lambda^{\deg p}$ only makes q a monic polynomial.

Suppose r is the minimal polynomial of λT , then $s(z) = \frac{1}{\lambda^{\deg p}} r(\lambda z)$ and $s(T) = \frac{1}{\lambda^{\deg p}} r(\lambda T) = 0$, thus $\deg s = \deg r = \deg p = \deg q$. $\square$

Exercise 3.19. Let V finite and $T \in \mathcal{L}(V)$. Let $\mathcal{E} \subseteq \mathcal{L}(V)$ a subspace, defined by

$$\mathcal{E} = \{ q(T) \mid q \in \mathcal{P}(F) \}$$

.

Show that $\dim \mathcal{E}$ is equal to the degree of the minimal polynomial of T .

Proof. We can see $I, T, T^2, \dots, T^{\deg p - 1}$ is linear independent (since p is the minimal polynomial of T) where p is the minimal polynomial of T . For any $q \in \mathcal{P}(F)$ where $\deg q \geq \deg p$, then $q = sp + r$ where $s, r \in \mathcal{P}(F)$ and $\deg r < \deg p$, therefore $q(T) = s(T)p(T) + r(T) = r(T) \in \text{span}(I, T, T^2, \dots, T^{\deg p - 1})$. $\square$

Exercise 3.20. let $T \in \mathcal{L}(F^4)$, which eigenvalues are 3, 5, 8. Show that $(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0$.

Proof. Suppose p is the minimal polynomial of T , then $p(z) = c(z - 3)(z - 5)(z - 8)q(z)$ since 3, 5, 8 are the eigenvalue of T , thus the zeros of p . Note that $\deg q \leq 1$ since $\deg p \leq \dim F^4 = 4$. Since there is no other eigenvalue (thus zero) than 3, 5, 8, q is either 1 or one of $z - 3$, $z - 5$, $z - 8$, thus $(z - 3)^2(z - 5)^2(z - 8)^2$ is polynomial multiple of p , therefore $(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0$. $\square$

Exercise 3.21. Let V finite and $T \in \mathcal{L}(V)$. Show that the degree of the minimal polynomial of T caps at $1 + \dim \text{range } T$.

Proof. IDK $\square$

Exercise 3.22. Let V finite and $T \in \mathcal{L}(V)$. Show that T is invertible $\iff I \in \text{span}(T, T^2, \dots, T^{\dim V})$.

Proof.

- ($\Rightarrow$) Suppose T is invertible, then the minimal polynomial p of T satisfies $p(0) \neq 0$ (since $p(0) = 0$ implies 0 is a eigenvalues of T). We know $\deg p \leq \dim V$, thus there is a linear combination of $T, T^2, \dots, T^{\dim V}$ that is equal to a scalar multiple of I , therefore $I \in \text{span}(T, T^2, \dots, T^{\dim V})$.
- ($\Leftarrow$) Suppose $I = \lambda_1 T + \lambda_2 T^2 + \dots + \lambda_{\dim V} T^{\dim V}$, then $I = T(\lambda_1 I + \lambda_2 T + \dots + \lambda_{\dim V} T^{\dim V - 1}) = (\lambda_1 I + \lambda_2 T + \dots + \lambda_{\dim V} T^{\dim V - 1})T$, thus T is invertible.

$\square$

Exercise. Let V a vector space and $T \in \mathcal{L}(V)$, $v, Tv, \dots, T^k v$ a list of linear independent vectors but $v, Tv, \dots, T^{k+1} v$ isn't. Show that $T^{k+i} v \in \text{span}(v, Tv, \dots, T^k v)$ for all $0 < i$.

Proof. Induction on i .

- Base($i = 1$): By assumption.
- Ind($i = i+1$): $T^{k+i+1} v = T(T^{k+i} v)$, since $T^{k+i} v \in \text{span}(v, Tv, \dots, T^k v)$, thus it can be write as a linear combination of $v, Tv, \dots, T^k v$, say $T(\lambda_0 v + \lambda_1 Tv + \dots + \lambda_k T^k v)$, then $\lambda_0 Tv + \lambda_1 T^2 v + \dots + \lambda_k T^{k+1} v \in \text{span}(v, Tv, \dots, T^k v)$ since $T^{k+1} v \in \text{span}(v, Tv, \dots, T^k v)$.

$\square$

Exercise 3.23. Let V finite and $T \in \mathcal{L}(V)$. Let $n = \dim V$, show that for any $v \in V$, $\text{span}(v, Tv, \dots, T^{n-1}v)$ is invariant under T .

Proof. Note that the list $v, Tv, \dots, T^{n-1}v$ has length $n = \dim V$, thus for the list $v, Tv, \dots, T^n v$ is linear dependent, thus $T^n v$ must be a linear combination of $v, Tv, \dots, T^{n-1}v$.

- If $v, Tv, \dots, T^{n-1}v$ is linear dependent, then $T^n v \in \text{span}(v, Tv, \dots, T^{n-1}v)$ (by our lemma exercise).
- Otherwise, the list $v, Tv, \dots, T^n v$ is linear dependent while $v, Tv, \dots, T^{n-1}v$ isn't, therefore $T^n v$ is a linear combination of $v, Tv, \dots, T^{n-1}v$.

□

Theorem 3.29. $q(T) = 0 \iff q$ is a polynomial multiple of the minimal polynomial of T .

Proof. • $(\Rightarrow)$ Let p the minimal polynomial of T , consider $q = sp + r$ where $\deg r < \deg p$, we may suppose $r \neq 0$. Then $0 = q(T) = s(T)p(T) + r(T) = r(T)$, which contradict to the assumption that p is the minimal polynomial of T .

- $(\Leftarrow)$ Trivial.

□

Exercise 3.25. Let V finite, $T \in \mathcal{L}(V)$, subspace $U \subseteq V$ is invariant under T .

- Show that the minimal polynomial of T is polynomial multiple of the minimal polynomial of T/U .
- Show that

$$(\text{the minimal polynomial of } T|_U) \times (\text{the minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial of T .

Proof. • Let p the minimal polynomial of T , then $p(T/U)(v + U) = p(T)v + U = 0 + U$ for any $v + U \in V/U$, thus $p(T/U) = 0$, therefore p is a polynomial multiple of the minimal polynomial of T/U .

- Let p the minimal polynomial of $T|_U$ and q the minimal polynomial of T/U . Then $(pq)(T)v = (p(T)q(T))v = p(T)(q(T)v)$ where $q(T)v \in U$, thus $p(T)(q(T)v) = 0$.

□

Exercise 3.26. Let V finite, $T \in \mathcal{L}(V)$, U is invariant under T . Show that the set of eigenvalues of T is equal to the union of eigenvalues of $T|_U$ and T/U .

Proof. This theorem separate the eigenvalues into two parts: eigenvectors in U and eigenvectors not in U (may have intersection).

- ($\subseteq$) For any $Tv = \lambda v$ where non-zero $v \in V$. If $v \in U$, then $T|_U(v) = Tv = \lambda v$. If $v \notin U$, then $(T/U)(v+U) = Tv+U = \lambda v+U = \lambda(v+U)$.
- ($\supseteq$) For any $T|_U(v) = \lambda v$, we have $T|_U(v) = Tv = \lambda v$. The case of T/U is proven in Exercise 5.38 of E5A.

□

We will use this conclusion several times, so we prove it first.

Exercise. Let p, q two non-constant **monic** polynomial and $p = sq$, $q = tp$ where s, t two non-zero polynomial. Show that $p = q$.

Proof. We have $p = stp$, thus $st = 1$ and $\deg s = \deg t = 0$. Furthermore, we have $p = sq$ where p and q are monic, thus s must be 1, similar to t , hence $s = t = 1$ and $p = q$. □

Exercise 3.27. Let $F = R$ and V finite and $T \in \mathcal{L}(V)$. Show that the minimal polynomial of T_C is equal to the T one.

Proof. Let p the minimal polynomial of T and q the minimal polynomial of T_C . We have:

$$\begin{aligned}
 & p(T_C)(v + iu) \\
 &= p(T)v + ip(T)u \\
 &= 0v + i0u \\
 &= 0
 \end{aligned}$$

and

$$\begin{aligned} & q(T)(v) \\ &= q(T_C)(v + i0) \\ &= 0 \end{aligned}$$

thus $p = sq$ and $q = tp$ where s, t are non-zero polynomials, therefore $p = q$. $\square$

Exercise 3.28. Let V finite and $T \in \mathcal{L}(V)$. Show that the minimal polynomial of $T' \in \mathcal{L}(V')$ is equal to the T one.

Proof. Let p the minimal polynomial of T and p' the minimal polynomial of T' .

For any $\varphi \in V'$ and $v \in V$, we have $p(T')(\varphi)(v) = \varphi(p(T)v) = \varphi 0 = 0$ (since φ is linear), thus $p(T')(\varphi) = 0$, therefore $p(T') = 0$.

For any $v \in V$, $p'(T)v = \varphi_1(p'(T)v)v_1 + \cdots = p'(T')(\varphi)(v) + \cdots = 0$, where $v_0, \cdots, v_{\dim V-1}$ is a basis of V and $\varphi_0, \cdots, \varphi_{\dim V-1}$ is a dual basis. Thus $p'(T) = 0$.

Hence, p and p' are polynomial multiple to each other, therefore $p = p'$. $\square$

Exercise 3.29. Let V finite, $T \in \mathcal{L}(V)$. Show that $\mathcal{M}(T)$ is upper-triangular for some basis of $V \iff \mathcal{M}(T')$ is upper-triangular for some basis of V' .

Proof. This follows that T and T' have the same minimal polynomial. See Exercise 5.28 in E5B. $\square$

Exercise 3.30. Prove or disprove: $T \in \mathcal{L}(V)$ and $\mathcal{M}(T^2)$ is upper-triangular for some basis of V , then $\mathcal{M}(T)$ is upper-triangular for some basis of V (not necessary the same as the $\mathcal{M}(T^2)$ one).

Proof. WoBuHui. $\square$

Exercise 3.31. Let A, B are upper-triangular matrices with same size, the diagonal of A is $\alpha_0, \cdots, \alpha_{n-1}$ and the diagonal of B is $\beta_0, \cdots, \beta_{n-1}$. Show that

- $A + B$ is upper-triangular and the diagonal is $\alpha_0 + \beta_0, \cdots, \alpha_{n-1} + \beta_{n-1}$.
- AB is upper-triangular and the diagonal is $\alpha_0\beta_0, \cdots, \alpha_{n-1}\beta_{n-1}$.

Proof.

- Trivial.
- Take the standard basis of F^n , we have $Bv_i \in \text{span}(v_0, \dots, v_i)$ and then $A(Bv_i) \in \text{span}(v_0, \dots, v_i)$ since both A and B are upper-triangular, thus AB is upper-triangular. For the diagonal, we know $AB_{i,i} = A_{i,-}B_{-,i}$, however, components before i -th of $A_{i,-}$ are 0 and components since i -th of $B_{-,i}$ are 0, therefore $AB_{i,i} = A_{i,i}B_{i,i} = \alpha_i\beta_i$.

□

Exercise 3.32. Let $T \in \mathcal{L}(V)$ invertible, and $\mathcal{M}(T)$ with respect to the basis $v_0, \dots, v_{n-1}$ of V is upper-triangular, while the diagonal is $\lambda_0, \dots, \lambda_{n-1}$. Show that $\mathcal{M}(T^{-1})$ with respect to that basis is also upper-triangular, and the diagonal is $\frac{1}{\lambda_0}, \dots, \frac{1}{\lambda_{n-1}}$.

Proof. For any $i = 1, \dots, n$, $\text{span}(v_0, \dots, v_{i-1})$ is invariant under T , thus it is invariant under T^{-1} since T^{-1} is the inverse of T .

For the diagonal, $TT^{-1} = I$, which diagonal is $1, \dots, 1$, which is equal to $\lambda_0\beta_0, \dots, \lambda_{n-1}\beta_{n-1}$ where β_i is the diagonal of T^{-1} . Thus $\beta_i = \frac{1}{\lambda_i}$. □

Exercise 3.33. Give an example that T an invertible operator, where the diagonal of $\mathcal{M}(T)$ is all 0.

Proof.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

□

Exercise 3.34. Give an example that T an singular operator, where the diagonal of $\mathcal{M}(T)$ is all non-zero.

Proof.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

□

Exercise 3.35. Let $F = C$ and V finite, and $T \in \mathcal{L}(V)$. Show that $k = 1, \dots, \dim V$, then there is a k -dimension subspace of V that is invariant under T .

Proof. If $F = C$, then $\mathcal{M}(T)$ is upper-triangular for some basis of V . Thus $\text{span}(v_0, \dots, v_{k-1})$ is invariant under T where v_i is such basis. $\square$

Exercise 3.36. Let V finite and $T \in \mathcal{L}(V)$ and $v \in V$. Show that:

- There is a unique monic polynomial p_v with minimal degree such that $p_v(T)v = 0$
- Show that the minimal polynomial of T is polynomial multiple of p_v .

Proof.

- $p(T)v = 0$, therefore we only need to show the uniqueness. Let s, t a monic polynomial with minimal degree such that $s(T)v = t(T)v = 0$, then $(s - t)(T)v = 0$, therefore $s = t$, otherwise there is a polynomial $s - t$ with lower degree such that $(s - t)(T)v = 0$.
- We divide p by p_v , then $p = sp_v + r$ where $s, r \in \mathcal{P}(F)$ and $\deg r < \deg p_v$. Therefore $r = 0$, otherwise r is a lower polynomial such that $r(T)v = 0$, which contradict the property of p_v . Thus $p = sp_v$.

$\square$

Exercise 3.37. Let V finite and $T \in \mathcal{L}(V)$, and non-zero $v \in V$ such that $T^2 + 2Tv + 2v = 0$. Show that

- If $F = R$, then $\mathcal{M}(T)$ is **NOT** upper-triangular for all basis of V .
- If $F = C$, then the diagonal of upper-triangular $\mathcal{M}(T)$ contains $-1 + i$ and $-1 - i$.

Proof.

- Note that $p_v(z) = z^2 + 2z + 2$ is a minimal polynomial of Tv , it is minimal since p_v has no zero, therefore cannot have lower degree.

Then the minimal polynomial p of T is a polynomial multiple of p_v , thus p is **NOT** in form of $(z - \lambda_0) \cdots (z - \lambda_{n-1})$ since p_v has no zero, thus there is no upper-triangular matrix for T for any basis of V .

- $-1 + i$ and $-1 - i$ are two zeros of p_v , thus are zeros of p , therefore are in the diagonal.

$\square$

Exercise 3.38. Let B square matrix with complex elements. Show that there is a square matrix A with complex elements such that $A^{-1}BA$ is a upper-triangular matrix.

Proof. We can find an operator T such that its matrix is B with respect to the standard basis. Then we can find a basis such that $\mathcal{M}(T)$ with respect to such basis is upper-triangular since B is complex. Then $A = \mathcal{M}(I, \text{standard basis}, \text{upper-triangular basis})$, and $A^{-1}BA$ is upper-triangular, this is the change-of-basis formula. $\square$

Exercise 3.39. Let $T \in \mathcal{L}(V)$ and $v_0, \dots, v_{n-1}$ a basis of V , show that the following statements are equivalent:

- the matrix of T with respect to $v_0, \dots, v_{n-1}$ is lower-triangular.
- For any $k = 1, \dots, n$, $\text{span}(v_{k-1}, \dots, v_{n-1})$ is invariant under T .
- For any $k = 1, \dots, n$, $Tv_{k-1} \in \text{span}(v_{k-1}, \dots, v_{n-1})$.

Proof. The proof is similar to the upper-triangular one.

- (1) $\Rightarrow$ (2) For any $i \leq j$, $Tv_{j-1} \in \text{span}(v_{j-1}, \dots, v_{n-1}) \subseteq \text{span}(v_{i-1}, \dots, v_{n-1})$, thus $\text{span}(v_{k-1}, \dots, v_{n-1})$ is invariant under T .
- (2) $\Rightarrow$ (3) Trivial.
- (3) $\Rightarrow$ (1) Basically the definition.

$\square$

Exercise 3.40. Let $F = C$ and V finite. Show that $T \in \mathcal{L}(V)$, then $\mathcal{M}(T)$ is lower-triangular with respect to some basis of V .

Proof. Consider the dual map T' , we know there is a basis of V' such that $\mathcal{M}(T')$ is upper-triangular, then $\mathcal{M}(T') = \mathcal{M}(T)^T$ which means $\mathcal{M}(T)^T$ is a upper-triangular, thus $(\mathcal{M}(T)^T)^T = \mathcal{M}(T)$ is lower-triangular. $\square$

Exercise 3.41. Let V finite and the matrix of $T \in \mathcal{L}(V)$ is upper-triangular with respect to some basis of V , and $U \subseteq V$ is invariant under T . Show that

- The matrix of $T|_U$ is upper-triangular with respect to some basis of U .

- The matrix of T/U is upper-triangular with respect to some basis of V/U .

Proof.

- Since $\mathcal{M}(T)$ is upper-triangular, then the minimal polynomial of T is in form of $p(z) = (z - \lambda_0) \cdots (z - \lambda_{n-1})$. Then $p(T|_U) = 0$, thus p is polynomial multiple of the minimal polynomial q of $T|_U$. therefore q is also in form of $(z - \lambda_0) \cdots (z - \lambda_{k-1})$. Thus there is a basis of U such that the matrix of $T|_U$ is upper-triangular.
- Let q the minimal polynomial of T/U , and p the minimal polynomial of T , then p is polynomial multiple of q (see Exercise 5.25 in E5B). Then follow the same step as last proof.

□

Exercise 3.42. Let V finite, $T \in \mathcal{L}(V)$, $U \subseteq V$ invariant under T , $\mathcal{M}(T|_U)$ is upper-triangular for some basis of U , $\mathcal{M}(T/U)$ is upper-triangular for some basis of V/U . Show that $\mathcal{M}(T)$ is upper-triangular for some basis of V .

Proof. We will use the conclusion of Exercise 5.25 in E5B:

$$st = (\text{the minimal polynomial of } T|_U) \times (\text{the minimal polynomial of } T/U)$$

is a polynomial multiple of the minimal polynomial p of T . Thus st is in form of $(z - \lambda_0) \cdots (z - \lambda_{n-1})$ since both $\mathcal{M}(T|_U)$ and $\mathcal{M}(T/U)$ are upper-triangular for some basis, therefore p is also in form of $(z - \lambda_0) \cdots (z - \lambda_{k-1})$, hence $\mathcal{M}(T)$ is upper-triangular for some basis of V . □

Exercise 3.43. Let V a vector space over $\mathbb{C}$ and $T \in \mathcal{L}(V)$. Show that

- $T^4 = I$ implies T is diagonalizable.
- $T^4 = T$ implies T is diagonalizable.
- Give an example that $T \in \mathcal{L}(C^2)$ such that $T^4 = T^2$ while T is not diagonalizable.

Proof.

- $T^4 = I$ implies $p(z) = z^4 - 1$ and $p(T) = 0$, then $p(z) = (z + i)(z - i)(z + 1)(z - 1)$ where $i, -i, 1, -1$ are distinct to each others, and p is polynomial multiple of the minimal polynomial of T , thus T is diagonalizable.
- $T^4 = T$ implies $p(z) = z^4 - z$ and $p(T) = 0$, then $p(z) = (z - 0)(z^3 - 1) = (z - 0)(z - (\cos(120^\circ) + i \sin(120^\circ)))(z - (\cos(120^\circ - i \sin(120^\circ))))(z - 1)$, thus T is diagonalizable.
- Maybe $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$? I have no idea why C^2 .

□

Exercise 3.3. Let V finite and $T \in \mathcal{L}(V)$ Show that T is diagonalizable, then $V = \text{null } T \oplus \text{range } T$.

Proof. Let $v \in \text{null } T \cap \text{range } T$, then $Tv = 0$ and $Tv = T(c_0v_0 + \cdots + c_{n-1}v_{n-1}) = \lambda_0c_0v_0 + \cdots + \lambda_{n-1}c_{n-1}v_{n-1}$ where λ_i are the numbers in the diagonal of the matrix of T . Since $v_0, \cdots, v_{n-1}$ is linear independent, then $\lambda_0c_0, \cdots, \lambda_{n-1}c_{n-1}$ is all zero, if $\lambda_0, \cdots, \lambda_{n-1}$ is not all zero, then $v = 0$, which means $\text{null } T + \text{range } T$ is a direct sum and $\dim V = \text{null } T + \text{range } T$, thus $V = \text{null } T \oplus \text{range } T$. If $\lambda_0, \cdots, \lambda_{n-1}$ is all zero, then $T = 0$, thus $V = \text{null } T \oplus \{0\}$. □

Exercise 3.5. Let V finite vector space over $\mathbb{C}$ and $T \in \mathcal{L}(V)$. Show that T is diagonalizable $\iff V = \text{null}(T - \lambda I) \oplus \text{range}(T - \lambda I)$ for any $\lambda \in \mathbb{C}$.

Proof.

- For any $\lambda \in \mathbb{C}$, $T - \lambda I$ is also diagonalizable since both T and $-\lambda I$ are diagonalizable, thus $V = \text{null}(T - \lambda I) \oplus \text{range}(T - \lambda I)$.
- We induction on the dimension of V

Base($\dim V = 1$): Clearly T is diagonalizable.

Ind($\dim V = n + 1$): Since T is an operator of finite complex vector space, then there is $\lambda \in \mathbb{C}$ an eigenvalue of T , thus $\text{null}(T - \lambda I) = E(\lambda, T)$, also we have $\text{range}(T - \lambda I)$ is invariant under T (since $\text{range } p(T)$ is invariant under T for any $p \in \mathcal{P}(F)$). We may define $U = \text{range}(T - \lambda I)$, then $T|_U$ is an operator on U , which has lower

dimension. Furthermore, for any $\alpha \in \mathbb{C}$, $\text{null}(T|_U - \alpha I) \subseteq \text{null}(T - \alpha I)$ and $\text{range}(T|_U - \alpha I) \subseteq \text{range}(T - \alpha I)$, also $\text{null}(T - \alpha I) \cap \text{range}(T - \alpha I) = \{0\}$, thus $\text{null}(T|_U - \alpha I) \cap \text{range}(T|_U - \alpha I) = \{0\}$, therefore $U = \text{null}(T|_U - \alpha I) \oplus \text{range}(T|_U - \alpha I)$. Hence $T|_U$ is diagonalizable by the induction hypothesis, then $U = E(\lambda_1, T|_U) \oplus \cdots \oplus E(\lambda_{n-1}, T|_U)$ where $E(\lambda_i, T|_U) \subseteq E(\lambda_i, T)$, also, $V = E(\lambda, T) \oplus U$, therefore $E(\lambda_i, T|_U) = E(\lambda_i, T)$ otherwise the dimension doesn't match.

Now, $V = E(\lambda, T) \oplus E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_{n-1}, T)$, thus T is diagonalizable.

□

Exercise 3.6. Let $T \in \mathcal{L}(F^5)$ and $\dim E(8, T) = 4$, show that $T - 2I$ or $T - 6I$ is invertible.

Proof. Basically it says that $\dim F^5 = 5$ and $\dim E(8, T) = 4$, therefore T can only have one another eigenvalue, thus T can not have both eigenvalue 2 and 6, therefore one of $T - 2I$ and $T - 6I$ is invertible. □

Exercise 3.7. Let $T \in \mathcal{L}(V)$ and T invertible. Show that

$$E(\lambda, T) = E\left(\frac{1}{\lambda}, T^{-1}\right)$$

for all $\lambda \in F$ where $\lambda \neq 0$.

Proof. For any $\lambda \in F$ where $\lambda \neq 0$, if

- λ is an eigenvalue of T , then $\frac{1}{\lambda}$ is an eigenvalue of T^{-1} . For any $v \in E(\lambda, T)$, then $T^{-1}v = T^{-1}(\frac{1}{\lambda}Tv) = \frac{1}{\lambda}T^{-1}(Tv) = \frac{1}{\lambda}v$ and vice versa
- λ is not an eigenvalue of T , then $\frac{1}{\lambda}$ is not an eigenvalue of T^{-1} , therefore $E(\lambda, T) = E(\frac{1}{\lambda}, T^{-1}) = \{0\}$.

□

Exercise 3.10. Find $R, T \in \mathcal{L}(F^4)$ with eigenvalues 2, 6, 7 only, that there is no $S \in \mathcal{L}(F^4)$ such that $R = S^{-1}TS$.

Proof. Let $R \in \mathcal{L}(F^4)$ such that $\dim E(2, R) = 2$ and $R \in \mathcal{L}(F^4)$ such that $\dim E(6, R) = 2$. Then for any $S \in \mathcal{L}(F^4)$ such that $R = S^{-1}TS$, we have $R - 2I = S^{-1}TS - 2I = S^{-1}TS - 2S^{-1}IS = S^{-1}(T - 2I)S$. Then $\text{null}(R - 2I) = \text{null}(S^{-1}(T - 2I)S)$, it is easy to see that $\dim \text{null}(S^{-1}(T - 2I)S) = \dim \text{null}(T - 2I) = \dim E(2, T)$, however $\dim E(2, T) = 1$ and $\dim E(2, R) = 2$, therefore no such S . $\square$

Exercise 3.11. Find $T \in \mathcal{L}(\mathbb{C}^3)$ such that 6, 7 are eigenvalues of T , and T is not diagonalizable.

Proof. That means we need to find a T such that which minimal polynomial is $p(z) = (z - 6)(z - 7)^2$ or $(z - 6)^2(z - 7)$, we will find one for the former one. The formula reminds us that there is $w \in \mathbb{C}^3$ such that $(T - 7I)w \neq 0$ but $(T - 7I)^2w = 0$, which means $(T - 7I)w \in E(7, T)$. We may let $w = (0, 0, 1)$ and $(T - 7I)w = (0, 1, 0)$, which is much simple. Then $T(0, 0, 1) - (0, 0, 7) = (0, 1, 0)$ gives us $T(0, 0, 1) = (0, 1, 7)$ and we can get:

$$\mathcal{M}(T) = \begin{bmatrix} 6 & & \\ & 7 & 1 \\ & & 7 \end{bmatrix}$$

with respect to the standard basis of $\mathbb{C}^3$.

Clearly for $p(z) = (z - 6)(z - 7)^2$, we have $p(T) = 0$, and 6, 7 are eigenvalues of T , thus we only need to show that $q(z) = (z - 6)(z - 7)$ doesn't make $q(T) = 0$ (cause $\deg p = 3$ and $\deg q = 2$, where the minimal polynomial is a polynomial multiple of q). Then $(T - 6I)(T - 7I)(0, 0, 1) = (T - 6I)(0, 1, 0) \neq 0$ since $(0, 1, 0) \notin E(6, T)$. Thus T has eigenvalue 6, 7 and cannot be diagonalized. $\square$

Exercise 3.12. Let $T \in \mathcal{L}(\mathbb{C}^3)$ such that 6, 7 are eigenvalues of T , and T is not diagonalizable. Show that there is $(z_0, z_1, z_2) \in \mathbb{C}^3$ such that $T(z_0, \dots, z_2) = (6 + 8z_0, 7 + 8z_1, 13 + 8z_2)$.

Proof. Before proving, we can verify the previous proof, we can see: $T(-3, -7, -20)$ holds (by solving equation like $6 + 8z_0 = 6z_0$).

And I can't prove it (maybe i can, but 2 complicate.) $\square$

Exercise 3.13. Let A is diagonal matrix with distinct element in diagonal, and matrix B has the same size as A . Show that $AB = BA \iff B$ is diagonal matrix.

Proof. ($\Leftarrow$) is trivial, since elements in a field is communitive on multiplication.

We can see i -th line of AB is α_i times i -th line of B , and the i -th column of BA is α_i times i -th column of B .

Thus for any $i, j = 0, \dots, n-1$, we have $(AB)_{i,j} = \alpha_i B_{i,j}$ and $(BA)_{i,j} = \alpha_j B_{i,j}$, thus $\alpha_i B_{i,j} = \alpha_j B_{i,j}$. If $B_{i,j} = 0$, then the proof is complete, otherwise $\alpha_i = \alpha_j$, while the elements in diagonal of A are distinct, thus $i = j$.

Therefore, elements that is not in the diagonal of B is 0, hence B is a diagonal matrix. $\square$

Exercise 3.14. • Find V a finite vector space over $\mathbb{C}$ and $T \in \mathcal{L}(V)$, such that T^2 is diagonalizable but T isn't.

- Let $F = \mathbb{C}$ and k a positive integer, show that T is diagonalizable $\iff T^k$ is diagonalizable.

Proof.

- $V = \mathbb{C}^2$ and $T(x, y) = (y, 0)$, which matrix is (with respect to the standard basis):

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

and $p(T) = T^2 = 0$ is the minimal polynomial of T .

- ($\Rightarrow$) if T is diagonalizable obviously T^k is diagonalizable.
 ($\Leftarrow$) if T^k is diagonalizable, let p the minimal polynomial of T^k . Then let $q(z) = (z^k - \lambda_0) \cdots (z^k - \lambda_{m-1})$ where $\lambda_0, \dots, \lambda_{m-1}$ are distinct elements in the diagonal of T^k , therefore $q(T) = (T^k - \lambda_0) \cdots (T^k - \lambda_{m-1}) = 0$, hence q is polynomial multiple of the minimal polynomial of T .

Note that for complex number c , $z^k - c$ have k different zeros, then $q(z)$ consists of k^k term like $z - \lambda_i$ where λ_i are distinct to each others. Thus the minimal polynomial of T is also in a similar form such that T is diagonalizable. $\square$

Exercise 3.15. Let V finite vector space over $\mathbb{C}$, $T \in \mathcal{L}(V)$, and p is the minimal polynomial of T . Show that the following statements are equivalent:

- T is diagonalizable

- There is no $\lambda \in \mathbb{C}$ such that p is polynomial multiple of $(z - \lambda)^2$.
- p and p' share no zero.
- The greatest common divisor (gcd) of p and p' is 1 (in other words, they are coprime).

Proof.

- (1) $\Rightarrow$ (2) Trivial, since p must in form of $(z - \lambda_0) \cdots (z - \lambda_{n-1})$ where $\lambda_0, \dots, \lambda_{n-1}$ are distinct.
- One idea is that p and p' share the same zero means $z - \lambda$ in both p and p' , thus there must be $(z - \lambda)^2$ or higher in p so that $(z - \lambda)$ in p' . But I can't give a formal prove, maybe FIXME later.
- (3) $\Rightarrow$ (2) If p is polynomial multiple of $(z - \lambda)^2$, then $p(z) = (z - \lambda)^2 q(z)$, thus $p'(z) = (z - \lambda)^2 q'(z) + 2(z - \lambda)q(z)$, thus λ is zero of p and p' , which contradict our assumption.
- (3) $\Rightarrow$ (4) Let d is the common divisor of p and p' . The zero of d is also the zero of p and p' , however, we know there is no zero for d since our assumption (since our vector space is finite and over $\mathbb{C}$). That means $\deg d = 1$, also, p is monic polynomial, and $d(z)$ is the gcd of the first coefficient of p and p' , which is $\gcd(1, \lambda) = 1$.

In fact (3) $\iff$ (4), (3) $\Leftarrow$ (4) is much trivial.

□

Exercise 3.16. Let $T \in \mathcal{L}(V)$ diagonalizable, $\lambda_0, \dots, \lambda_{n-1}$ are distinct eigenvalues of T . Show that $U \subseteq V$ a subspace is invariant under $T \iff$ there is $U_0, \dots, U_{n-1} \subseteq V$ such that $U_{k-1} \subseteq E(\lambda_{k-1}, T)$ and $U = U_0 \oplus \cdots \oplus U_{n-1}$.

Proof.

- ($\Rightarrow$) We can see $T|_U$ is an operator on U , consider the minimal polynomial p of $T|_U$, then $p(T|_U) = 0$, therefore p is polynomial multiple of the minimal polynomial of $T|_U$, also p is in form of $(z - \lambda_0) \cdots (z - \lambda_{n-1})$ where λ_i are distinct, thus the minimal polynomial of $T|_U$ is in a similar form, therefore $T|_U$ is diagonalizable with eigenvalues $\lambda_0, \dots, \lambda_{m-1}$, note that $T|_U$ may have less eigenvalues. Then $U = E(\lambda_0, T|_U) \oplus \cdots \oplus$

$E(\lambda_{m-1}, T|_U) \oplus E(\lambda_m, T|_U) \oplus \cdots \oplus E(\lambda_{n-1}, T|_U)$ where $E(\lambda_{i-1}, T|_U) \subseteq E(\lambda_{i-1}, T)$ for all $i = 1, \dots, m$ and $E(\lambda_{m+i-1}, T|_U) = \{0\} \subseteq E(\lambda_{m+i-1}, T)$ for all $i = 1, \dots, n - m$.

- Trivial, for any $u \in U_{k-1} \subseteq E(\lambda_{k-1}, T)$, we have $Tu = \lambda_{k-1}u$, thus U_{k-1} is invariant under T , therefore U is invariant under T , where $U = U_0 \oplus \cdots \oplus U_{n-1}$

□

Exercise 3.17. Let V finite, show that there is a basis which consists of diagonalizable operators for $\mathcal{L}(V)$.

Proof. Let $v_0, \dots, v_{n-1}$ a basis of V , then define $T_{i-1}(v) = \varphi_i(v)v_i$ where φ_i is dual basis of $v_0, \dots, v_{n-1}$. Then $T_0, \dots, T_{n-1}$ is linear independent with length n , therefore it is a basis of $\mathcal{L}(V)$.

T_i is diagonalizable since $p(z) = (z - 0)(z - 1)$ and $p(T) = 0$. □

Exercise 3.18. Let $T \in \mathcal{L}(V)$ is diagonalizable, and $U \subseteq V$ is invariant under T . Show that $T|_U$ is diagonalizable.

Proof. The minimal polynomial of T is polynomial multiple of the minimal polynomial of $T|_U$. □

Exercise 3.19. Prove or disprove: Let $T \in \mathcal{L}(V)$ and U is invariant under T , $T|_U$ and T/U is diagonalizable show that T is diagonalizable.

Proof. Unlike a similar statement about upper-triangular (see Exercise 5.13 in E5C), diagonalizable matrix requires that which minimal polynomial has no factor like $(z - \lambda)^2$.

Consider $T(x, y) = (x, x + y)$, which matrix is:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

.

Let $U = \text{span}((1, 0)) = E(1, T)$, then $T|_U$ and T/U is diagonalizable since U and V/U have dimension 1, however T is not diagonalizable since $p(z) = (z - 1)^2$ is the minimal polynomial of T , we have $(T - I)(0, 1) = (1, 1) - (0, 1) = (1, 0)$. □

Exercise 3.20. Let V finite and $T \in \mathcal{L}(V)$. Show that T is diagonalizable $\iff T'$ is diagonalizable

Proof. T and T' have the same minimal polynomial. □

Exercise 3.21. A fibonacci sequence $F_0, F_1, F_2, \dots$ is defined by:

$$F_0 = 0, F_1 = 1 \quad \text{and} \quad F_n = F_{n-2} + F_{n-1} \quad \forall n \geq 2.$$

Define $T \in \mathcal{L}(\mathbb{R}^2)$ by $T(x, y) = (y, x + y)$.

1. Show that $T^n(0, 1) = (F_n, F_{n+1})$ for any non-negative n .
2. Find the eigenvalues of T .
3. Find a basis of $\mathbb{R}^2$ which consists of eigenvectors of T .
4. Use (3) to calculate $T^n(0, 1)$ and conclude that:

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$$

for any non-negative n .

5. Use (4) to conclude that F^n is an integer that nearest to $\frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n$ for any non-positive n .

Proof.

- Induction on n .

$$\text{Base}(n = 0): T^0(0, 1) = (0, 1) = (F_0, F_1).$$

$$\text{Base}(n = 1): T^1(0, 1) = (1, 1) = (F_1, F_2).$$

$$\text{Ind}(n = n + 1): T^{n+1}(0, 1) = T(T^n(0, 1)) = T(F_n, F_{n+1}) = (F_{n+1}, F_n + F_{n+1}) = (F_{n+1}, F_{n+2}).$$

- Suppose $F(x, y) = (y, x + y) = \lambda(x, y)$ where $(x, y) \neq 0$, then:

$$\lambda x = y$$

$$\lambda y = x + y$$

therefore

$$\lambda^2 x = x + \lambda x$$

we may suppose $x \neq 0$, otherwise $x = y = 0$ then $(x, y) = 0$ gives us $\perp$, therefore

$$\lambda^2 = \lambda + 1$$

has solution $\lambda = \frac{1 \pm \sqrt{5}}{2}$.

- We denote $\frac{1+\sqrt{5}}{2}$ by λ_0 and $\frac{1-\sqrt{5}}{2}$ by λ_1 . Thus we have $E(\lambda_0, T) = \text{span}((1, \lambda_0), T)$ and $E(\lambda_1, T) = \text{span}((1, \lambda_1))$.
- First, by the change-of-basis formula, we have:

$$\mathcal{M}(T) = \begin{bmatrix} 1 & 1 \\ \lambda_0 & \lambda_1 \end{bmatrix} \begin{bmatrix} \lambda_0 & \\ & \lambda_1 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} \lambda_1 & 1 \\ -\lambda_1 & -1 \end{bmatrix}$$

therefore

$$\begin{aligned} \mathcal{M}(T^n(0, 1)) &= \begin{bmatrix} 1 & 1 \\ \lambda_0 & \lambda_1 \end{bmatrix} \begin{bmatrix} \lambda_0 & 0 \\ 0 & \lambda_1 \end{bmatrix}^n \frac{1}{\sqrt{5}} \begin{bmatrix} \lambda_1 & 1 \\ -\lambda_1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ \lambda_0 & \lambda_1 \end{bmatrix} \begin{bmatrix} \lambda_0^n & 0 \\ 0 & \lambda_1^n \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 1 \\ \lambda_0 & \lambda_1 \end{bmatrix} \begin{bmatrix} \lambda_0^n \\ -\lambda_1^n \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{bmatrix} \lambda_0^n - \lambda_1^n \\ \lambda_0^{n+1} - \lambda_1^{n+1} \end{bmatrix} \end{aligned}$$

where the first component is F_n and the second component is F_{n-1} , therefore:

$$F^n = \frac{1}{\sqrt{5}}(\lambda_0^n - \lambda_1^n)$$

where $\lambda = \frac{1 \pm \sqrt{5}}{2}$.

- We want to show that F^n is the nearest integer to $\frac{1}{\sqrt{5}}(\frac{1+\sqrt{5}}{2})^n$, that means we need to show:

$$\left| \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n \right| \leq \frac{1}{2}$$

Then $|\frac{1-\sqrt{5}}{2}| \leq 1$, thus we only need to show:

$$\begin{aligned} |\frac{1}{\sqrt{5}}(\frac{1-\sqrt{5}}{2})| &\leq \frac{1}{2} \\ |\frac{1-\sqrt{5}}{\sqrt{5}}| &\leq 1 \\ |1-\sqrt{5}| &\leq \sqrt{5} \end{aligned}$$

Obviously $\sqrt{5} > 2$ and $|1-\sqrt{5}| < 2$.

□

Exercise 3.22. Let $T \in \mathcal{L}(V)$ and A is an $n \times n$ matrix of T with respect to some basis of V . Show that

$$|A_{j,j}| > \sum_{\substack{k=1 \\ k \neq j}}^n |A_{j,k}|$$

implies T is invertible.

Proof. Gershgorin disk theorem says that any eigenvalue of T is in some gershgorin disk, therefore we can see 0 is not an eigenvalue of T , otherwise the inequality above becomes $\geq$. □

Exercise 3.23. Give an example that $S, T \in \mathcal{L}(F^4)$, such that there is a subspace that is invariant under S but not T , and another subspace that is invariant under T but not S .

Proof. $S(a, b, c, d) = (a, b, d, -c)$ and $T(a, c, -b, d)$, then $U = \text{span}((0, 0, 1, 0), (0, 0, 0, 1))$ and $W = \text{span}((0, 0, 1, 0), (0, 1, 0, 0))$, obviously U is invariant under S and W is invariant under T , and $T(0, 0, 1, 1) = (0, 1, -0, 1) \notin U$ and $S(0, 1, 1, 0) = (0, 1, 0, -1) \notin W$. □

Exercise 3.3. Let $S, T \in \mathcal{L}(V)$ and S, T commute. Let $p \in \mathcal{P}(F)$.

- Show that $\text{null } p(S)$ is invariant under T .
- Show that $\text{range } p(S)$ is invariant under T .

Proof.

- For any $v \in \text{null } p(S)$, we have $p(S)(Tv) = T(p(S)v) = T0 = 0$ (cause S, T commute), thus $Tv \in \text{null } p(S)$.
- For any $v \in \text{range } p(S)$, we have $Tv = T(p(S)w) = p(S)(Tw) \in \text{range } p(S)$.

□

Exercise 3.4. *Prove or disprove: Let A diagonal matrix and B upper-triangular matrix with same size as A , then A, B is commute.*

Proof. This can be disprove by Exercise 5.13 in E5D: A, B commute means B is diagonal matrix, as long as elements in the diagonal A are distinct. □

Exercise 3.5. *Show that a pair of operators in a finite vector space are commute $\iff$ their dual operators are commute.*

Proof.

$$\begin{aligned}
 ST = TS &\iff \mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(T)\mathcal{M}(S) \\
 &\iff (\mathcal{M}(T)^T \mathcal{M}(S)^T)^T = (\mathcal{M}(S)^T \mathcal{M}(T)^T)^T \\
 &\iff \mathcal{M}(T)^T \mathcal{M}(S)^T = \mathcal{M}(S)^T \mathcal{M}(T)^T \\
 &\iff \mathcal{M}(T')\mathcal{M}(S') = \mathcal{M}(S')\mathcal{M}(T') \\
 &\iff T'S' = S'T'
 \end{aligned}$$

□

Exercise 3.6. *Let V a non-zero, finite, complex vector space, and $S, T \in \mathcal{L}(V)$ are commute. Show that there is $\alpha, \lambda \in F$ such that*

$$\text{range}(S - \alpha I) + \text{range}(T - \lambda I) \neq V$$

Proof. The goal is find a common eigenvector of S, T , fortunately, there is common eigenvector for commute operators of non-zero, finite, complex vector. Thus α, λ are the eigenvalues of that common eigenvector. □

Exercise 3.7. *Let V complex vector space, $S \in \mathcal{L}(V)$ is diagonalizable, and $T \in \mathcal{L}(V)$ commutes with S . Show that there is a basis of V such that $\mathcal{M}(S)$ is diagonal and $\mathcal{M}(T)$ is upper-triangular.*

Proof. I guess V is finite. This proof basically a modified proof from book We induction on $\dim V$.

- Base($\dim V = 1$), trivial.
- Base($\dim V = n + 1$): We know S have at least one eigenvalue, then $V = \text{span}(v) \oplus W$ for some W .

Define $P(\alpha v + \beta w) = \beta w$ and $\hat{S}(w) = P(S(w))$ and $\hat{T}(w) = P(T(w))$ two operators in $\mathcal{L}(W)$. We will show that $\hat{S}, \hat{T}$ commute. $\hat{S}(\hat{T}w) = \hat{S}(P(Tw)) = \hat{S}(Tw - \alpha v) = P(STw - \alpha Sv) = P(TSw) - 0$, similarly, $\hat{T}(\hat{S}w) = P(STw)$, thus $\hat{S}\hat{T} = \hat{T}\hat{S}$.

We then will show that $\hat{S}$ is diagonalizable, suppose $v \in E(\lambda_0, S)$, then $V = E(\lambda_0, S) \oplus E(\lambda_1, S) \oplus \cdots \oplus E(\lambda_{m-1}, S)$, where $E(\lambda_0, S) = \text{span}(v) \oplus \text{span}(w) \oplus \cdots$, where $v, w, \cdots$ are basis of $E(\lambda_0, S)$, thus we have $W = (\text{span}(w) \oplus \cdots) \oplus E(\lambda_1, S) \oplus \cdots$. For any vector in $E(\lambda_k, S)$ is also an eigenvector of $\hat{S}$, and for any $v \in \text{span}(w) \oplus \cdots$, $\hat{S}(w + \cdots) = S(w + \cdots) = \lambda w + \cdots = \lambda(w + \cdots) \in \text{span}(w) \oplus \cdots$.

Then by induction hypothesis, there is a basis of V , say $v_1, \cdots, v_{n-1}$ that $\hat{S}$ is diagonal and $\hat{T}$ is upper-triangular. Then $v, v_1, \cdots, v_{n-1}$ is a basis of V where they are eigenvectors of S cause $\mathcal{M}(\hat{S})$ is diagonal and v is an eigenvector of S . Also we have $Tv = \lambda v$ as v is also an eigenvector of T , and $Tv_k = \hat{T}(v_k) \in \text{span}(v_1, \cdots, v_k) \subseteq \text{span}(v, v_1, \cdots, v_k)$.

□

Exercise 3.9. Let V finite, non-zero, complex vector space and $\mathcal{E} \subseteq \mathcal{L}(V)$, such that any pair in $\mathcal{E}$ is commute.

- Show that there is a vector in v such that it is eigenvector for all element in $\mathcal{E}$.
- Show that there is a basis of V such that any element in $\mathcal{E}$ is upper-triangular with respect to that basis.

Note that $\mathcal{E}$ can be infinite.

Proof.

- Take any two element in $\mathcal{E}$, say S, T , we know there is a common eigenvector v of S and T . Then $\text{span}(v)$ is invariant under all element of $\mathcal{E}$, and any element that restrict to $\text{span}(v)$ have an eigenvector, therefore v is the common eigenvector for all element in $\mathcal{E}$.

- Induction on $\dim V$:

Base($\dim V = 1$): trivial.

Ind($\dim V = n + 1$): Let v_0 the common eigenvector of all element of $\mathcal{E}$, then $V = \text{span}(v_0) \oplus W$ for some W . Define $P(\alpha v_0 + w) = w$, and $\hat{T}_i(w) = P(T_i(w))$ for all $T_i \in \mathcal{E}$. For any $T_i, T_j \in \mathcal{E}$, we have $\hat{T}_i \hat{T}_j(w) = \hat{T}_i(P(T_j)w) = \hat{T}_i(T_j w - \alpha v_0) = P(T_i T_j w - \alpha T_i v_0)$, recall that v_0 is the common eigenvector of all $T_k \in \mathcal{E}$, thus $P(T_i T_j w - \alpha T_i v_0) = P(T_i T_j w)$, similarly $\hat{T}_j \hat{T}_i w = P(T_j T_i w)$, therefore $\hat{T}_i, \hat{T}_j$ commute.

By induction hypothesis, we know there is a basis of W such that $\hat{T}_i$ is upper-triangular, say $v_1, \dots, v_{n-1}$. Then the basis $v_0, \dots, v_{n-1}$ makes all T_i upper-triangular, cause $T_i v_0 \in \text{span}(v_0)$ and $T_i v_j \in \text{span}(v_1, \dots, v_{n-1}) \subseteq \text{span}(v_0, \dots, v_{n-1})$ for any $j = 1, \dots, n - 1$.

□

Exercise 6.10. *Prove or disprove: If $v_0, \dots, v_{n-1} \in V$, then:*

$$\sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \langle v_j, v_k \rangle \geq 0$$

Proof. For any j , we have $\sum_{k=0}^{n-1} \langle v_j, v_k \rangle = \langle v_j, v_0 + \dots + v_{n-1} \rangle$, thus the equation is now $\langle v_0 + \dots + v_{n-1}, v_0 + \dots + v_{n-1} \rangle \geq 0$, which is trivial by definition. □

Exercise 6.11. *Let $S \in \mathcal{L}(V)$. Define $\langle \cdot, \cdot \rangle_1$ by*

$$\langle u, v \rangle_1 = \langle Su, Sv \rangle$$

for all $u, v \in V$. Show that $\langle \cdot, \cdot \rangle_1$ is an inner product over $V \iff S$ is injective.

Proof.

- Try to prove $\langle \cdot, \cdot \rangle_1$ is an inner product and see where we stuck without injective. We find that $\langle v, v \rangle_1 = 0 \iff v = 0$ stuck, as if S is not injective, then we have $\langle u-v, u-v \rangle_1 = \langle S(u-v), S(u-v) \rangle = \langle 0, 0 \rangle = 0$ where $u-v \neq 0$ and $Su = Sv$. Therefore S must be injective.

Although the proof is not *constructive*, it help us to find a constructive proof: Suppose $Su = Sv$, then $\langle u-v, u-v \rangle_1 = \langle Su-Sv, Su-Sv \rangle = 0$, thus $u-v=0$, therefore $u=v$.

- Positivity, Additivity, Homogeneity holds cause S is a linear map and $\langle \cdot, \cdot \rangle$ is an inner product, and Conjugate Symmetry holds cause $\langle \cdot, \cdot \rangle$ is an inner product. For Definiteness, $\langle v, v \rangle_1 = \langle Sv, Sv \rangle = 0$ implies $v=0$ cause S injective, and $\langle 0, 0 \rangle_1 = \langle S0, S0 \rangle = \langle 0, 0 \rangle = 0$.

□

Exercise 6.12.

- Show that $f((a, b), (c, d)) = |ac| + |bd|$ is not an inner product over $\mathbb{R}^2$.
- Show that $f((a, b, c), (x, y, z)) = ax + cz$ is not an inner product over $\mathbb{R}^3$.

Proof.

- $f((1, 1), (1, 1)) = 1 + 1$ and $f((-1, -1), (1, 1)) = 1 + 1$, then $f((1, 1) + (-1, -1), (1, 1)) = f((0, 0), (1, 1)) = 0 + 0 = 0$ while $f((1, 1) + (-1, -1), (1, 1)) = f((1, 1), (1, 1)) + f((-1, -1), (1, 1)) = 2 + 2 = 4$.
- $f((0, 1, 0), (0, 1, 0)) = 0$ but $(0, 1, 0) \neq 0$.

□

Exercise 6.13. Let $T \in \mathcal{L}(V)$ and $\|Tv\| \leq \|v\|$ for all $v \in V$. Show that $T - \sqrt{2}I$ is injective.

Proof. Suppose $T - \sqrt{2}I$ is not injective, then $Tv = \sqrt{2}v$ for some v , then $\|Tv\| = \|\sqrt{2}v\| = |\sqrt{2}| \|v\| \geq \|v\|$. Basically any eigenvalue with absolute value greater than 1 can make it. □

Exercise 6.14. Let V an inner product space over $\mathbb{R}$.

- Show that $\langle u+v, u-v \rangle = \|u\|^2 - \|v\|^2$.

- Show that $u + v \perp u - v$ if $\|u\| = \|v\|$.
- Use last conclusion to show that the diagonal of 菱形 are orthogonal.

Proof.

- $\langle u + v, u - v \rangle = \langle u, u \rangle + \langle v, u \rangle - \langle u, v \rangle - \langle v, v \rangle$, note that $\langle v, u \rangle = \langle u, v \rangle$ since V over $\mathbb{R}$, thus $\langle u + v, u - v \rangle = \langle u, u \rangle - \langle v, v \rangle = \|u\|^2 - \|v\|^2$.
- By $\uparrow$.
- We know 菱形 is a parallelogram that four sides have same length, thus $\|u\| = \|v\|$ and $u + v \perp u - v$, where $u + v$ and $u - v$ are two diagonal of 菱形.

□

Exercise 6.15. Let $u, v \in V$. Show that $\langle u, v \rangle = 0 \iff \|u\| \leq \|u + av\|$ for any $a \in F$.

Proof.

- ($\Rightarrow$) We will show that $\|u\| \leq \|u + av\|$ by $\|u\|^2 \leq \|u + av\|^2$ (recall that norm is always non-negative). $\|u + av\|^2 = \|u\|^2 + (a\|v\|)^2$
- ($\Leftarrow$) If $v = 0$, then $0 \perp u$ and the proof is complete, we assume $v \neq 0$. Let $w \in V$ such that $cv + w = u$ and $v \perp w$, then $\|u\|^2 = \|cv + w\|^2 = \|cv\|^2 + \|w\|^2$ (see theorem 6.13), thus $\|u\|^2 \geq \|w\|^2$, therefore $\|u\| \geq \|w\|$ where $w = u - cv$, therefore $\|w\| \leq \|u\|$, hence $\|u\| = \|w\|$. Then $\|u\|^2 = \|cv\|^2 = \|w\|^2$ is now $\|cv\|^2 = 0$, therefore $c = 0$ or $v = 0$, but $v \neq 0$, thus $c = \frac{\langle u, v \rangle}{\|v\|^2} = 0$ then $\langle u, v \rangle = 0$ and $u \perp v$.

□

Exercise 6.16. Let $u, v \in V$. Show that $\|au + bv\| = \|bu + av\|$ for any $a, b \in \mathbb{R} \iff \|u\| = \|v\|$.

Proof.

- We have $\|au + bv\|^2 = \langle au, au \rangle + \langle bv, bv \rangle + \langle au, bv \rangle + \langle bv, au \rangle$ and $\|bu + av\|^2 = \langle bu, bu \rangle + \langle av, av \rangle + \langle bu, av \rangle + \langle av, bu \rangle$. Note that $\langle au, bv \rangle = a\bar{b}\langle u, v \rangle = \bar{a}b\langle u, v \rangle = \langle bu, av \rangle$, similarly, $\langle bv, au \rangle = \langle av, bu \rangle$. Thus $\langle au, au \rangle + \langle bv, bv \rangle = \langle bu, bu \rangle + \langle av, av \rangle$, let $a = \sqrt{2}$ and $b = 1$, we have $\sqrt{2}^2\langle u, u \rangle + \langle v, v \rangle = \langle u, u \rangle + \sqrt{2}^2\langle v, v \rangle$ therefore $\langle u, u \rangle = \langle v, v \rangle$, thus $\|u\| = \|v\|$.
- $\|au + bv\|^2 - \|bu + av\|^2 = a^2\langle u, u \rangle + b^2\langle v, v \rangle + b^2\langle u, u \rangle + a^2\langle v, v \rangle = 0$.

□

Exercise 6.9. Let $\|u\| = \|v\| = 1$ and $\langle u, v \rangle = 1$. Show that $u = v$.

Proof. We know $\langle u, v \rangle = \langle v, u \rangle = 1$ and then $\langle u, v \rangle - \langle v, v \rangle = 1 - 1 = 0 = \langle u - v, v \rangle$ and $-\langle v, u \rangle + \langle u, u \rangle = -1 + 1 = 0 = \langle u - v, u \rangle$ therefore $\langle u - v, v \rangle - \langle u - v, u \rangle = \langle u - v, u - v \rangle = 0$, thus $u - v = 0$. □

Exercise 6.10. Let $u, v \in V$ and $\|u\|, \|v\| \leq 1$. Show that:

$$\sqrt{1 - \|u\|^2} \sqrt{1 - \|v\|^2} \leq 1 - |\langle u, v \rangle|$$

Proof. By Cauchy-Schwarz inequality, we have $1 - |\langle u, v \rangle| \geq 1 - \|u\| \|v\|$. We will show that $\sqrt{1 - \|u\|^2} \sqrt{1 - \|v\|^2} \leq 1 - \|u\| \|v\|$.

$$\begin{aligned} \sqrt{1 - \|u\|^2} \sqrt{1 - \|v\|^2} &\leq 1 - \|u\| \|v\| \\ (1 - \|u\|^2)(1 - \|v\|^2) &\leq (1 - \|u\| \|v\|)^2 \\ 1 - \|u\|^2 - \|v\|^2 + \|u\|^2 \|v\|^2 &\leq 1 - 2\|u\| \|v\| + \|u\|^2 \|v\|^2 \\ -\|u\|^2 - \|v\|^2 &\leq -2\|u\| \|v\| \\ \|u\|^2 + \|v\|^2 &\geq 2\|u\| \|v\| \\ \|u\|^2 - 2\|u\| \|v\| + \|v\|^2 &\geq 0 \\ (\|u\| - \|v\|)^2 &\geq 0 \end{aligned}$$

The second step is well since both sides are non-negative. □

Exercise 6.11. Find $u, v \in \mathbb{R}^2$, such that u a scalar multiple of $(1, 3)$ and v is perpendicular to $(1, 3)$ and $(1, 2) = u + v$.

Proof. 人话：找到一个关于 $(1, 3)$ 和垂直于 $(1, 3)$ 的向量的线性组合，使得它等于 $(1, 2)$ 。

We have $(3, -1) \perp (1, 3)$, then $x = \frac{7}{10}$ and $y = \frac{1}{10}$ is a solution of:

$$\begin{aligned} x + 3y &= 1 \\ 3x - y &= 2 \end{aligned}$$

Thus $u = \frac{7}{10}(1, 3)$ and $v = \frac{1}{10}(3, -1)$. □

Exercise 6.14. Let $v \in V$ and $v \neq 0$. Show that for any $u \in V$ and $\|u\| = 1$, then

$$\left\| v - \frac{v}{\|v\|} \right\| \leq \|v - u\|$$

equal only if $u = \frac{v}{\|v\|}$.

这段话说的是： v 穿过单位圆的点的向量是距离 v 最近的元素。

Proof. Let $v' = \frac{v}{\|v\|}$ and $w \in V$ where $w \perp v'$ such that $u = cv' + w$, we can see $|c| \leq 1$ by 勾股定理. Then

$$\begin{aligned} \|v - v'\| &\leq \|v - u\| \\ \|v - v'\|^2 &\leq \|v - u\|^2 \\ \langle v, v \rangle + \langle -v', -v' \rangle + \langle v, -v' \rangle + \langle -v', v \rangle &\leq \langle v, v \rangle + \langle -u, -u \rangle + \langle v, -u \rangle + \langle -u, v \rangle \\ \langle v, -v' \rangle + \langle -v', v \rangle &\leq \langle v, -u \rangle + \langle -u, v \rangle \\ -\langle v, v' \rangle - \langle v', v \rangle &\leq -\langle v, u \rangle - \langle u, v \rangle \\ (-1)(\langle v, v' \rangle + \overline{\langle v, v' \rangle}) &\leq (-1)(\langle v, cv' + w \rangle + \langle cv' + w, v \rangle) \\ \langle v, v' \rangle + \overline{\langle v, v' \rangle} &\geq \langle v, cv' + w \rangle + \langle cv' + w, v \rangle \\ \langle v, v' \rangle + \langle v, v' \rangle &\geq \langle v, cv' \rangle + \langle v, w \rangle + \langle cv', v \rangle + \langle w, v \rangle \\ 2\langle v, v' \rangle &\geq \langle v, cv' \rangle + \langle cv', v \rangle \\ 2\langle v, v' \rangle &\geq \bar{c}\langle v, v' \rangle + c\langle v', v \rangle \\ 2\langle v, v' \rangle &\geq \bar{c}\langle v, v' \rangle + c\overline{\langle v, v' \rangle} \\ 2\langle v, v' \rangle &\geq \bar{c}\langle v, v' \rangle + c\langle v, v' \rangle \\ 2 &\geq \bar{c} + c \\ 2 &\geq 2\operatorname{Re} c \\ 1 &\geq \operatorname{Re} c \end{aligned}$$

We have $\operatorname{Re} c \leq |c|$ and $|c| \leq 1$, thus $\operatorname{Re} c \leq 1$.

The inequality equals if $\operatorname{Re} c = 1$, which means $c = 1$ as $|c| = 1$ so the imaginary part is 0, thus $u = cv' + w = v' + w$ while $\|u\| = \|v'\| = 1$, which means $w = 0$, therefore $u = v'$. $\square$

Exercise 6.15. Let non-zero $u, v \in \mathbb{R}^2$. Show that:

$$\langle u, v \rangle = \|u\| \|v\| \cos \theta$$

where θ is the angle between u and v .

Proof. Note that the inner product of $\mathbb{R}^2$ is euclidian inner product, then let $u = a(\cos \alpha, \sin \alpha)$ and $v = b(\cos \beta, \sin \beta)$, thus

$$\begin{aligned} \langle u, v \rangle &= ab(\cos \alpha \cos \beta) + ab(\sin \alpha \sin \beta) \\ &= ab(\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\ &= ab(\cos \alpha \cos(-\beta) - \sin \alpha \sin(-\beta)) \\ &= ab(\cos(\alpha - \beta)) \\ &= \|u\| \|v\| (\cos(\alpha - \beta)) \end{aligned}$$

Recall that $\|u\| = a$ and $\|v\| = b$, since $\|(\cos \alpha, \sin \alpha)\| = 1$, similar to b . Also $\cos \theta = \cos -\theta$ and $\sin -\theta = -\sin \theta$. $\square$

Exercise 6.16. The angle between two vectors in $\mathbb{R}^n$ are hard to see when $n > 3$, thus define the angle between $u, v \in \mathbb{R}^n$ by:

$$\arccos \frac{\langle x, y \rangle}{\|x\| \|y\|}$$

, try explain why this requires Cauchy-Schwarz inequality.

Proof. Note that $\arccos$ is the inverse of $\cos$, thus $\arccos$ only defines at $[-1, 1]$, which means $|\frac{\langle x, y \rangle}{\|x\| \|y\|}| \leq 1$, Cauchy-Schwarz inequality says $|\langle x, y \rangle| \leq \|x\| \|y\|$, thus $|\frac{\langle x, y \rangle}{\|x\| \|y\|}| \leq 1$. $\square$

Exercise 6.20. Show that $u, v \in V$, then $|\|u\| - \|v\|| \leq \|u - v\|$.

Proof. We will show $|\|u\| - \|v\||^2 \leq \|u - v\|^2$, we can see that no matter $\|u\| - \|v\|$ greater or less than 0, $|\|u\| - \|v\||^2$ is always $\|u\|^2 + \|v\|^2 - 2\|u\|\|v\|$, thus

$$\begin{aligned}\|u\|^2 + \|v\|^2 - 2\|u\|\|v\| &\leq \langle u, u \rangle + \langle v, v \rangle + \langle u, -v \rangle + \langle -v, u \rangle \\ -2\|u\|\|v\| &\leq (-1)(\langle u, v \rangle + \langle v, u \rangle) \\ -2\|u\|\|v\| &\leq (-1)(\langle u, v \rangle + \overline{\langle u, v \rangle}) \\ -2\|u\|\|v\| &\leq (-2)\operatorname{Re}\langle u, v \rangle \\ \|u\|\|v\| &\geq \operatorname{Re}\langle u, v \rangle\end{aligned}$$

Recall that $|\langle u, v \rangle| \leq \|u\|\|v\|$ and $\operatorname{Re}\langle u, v \rangle \leq |\operatorname{Re}\langle u, v \rangle| \leq |\langle u, v \rangle|$. $\square$

Exercise 6.22. Let $u, v \in V$, show that

$$\|u + v\|\|u - v\| \leq \|u\|^2 + \|v\|^2$$

Proof. The left hand side is:

$$\begin{aligned}&\|u + v\|\|u - v\| \\ &= \sqrt{\langle u + v, u + v \rangle \langle u - v, u - v \rangle} \\ &= \sqrt{(\|u\|^2 + \|v\|^2 + \langle u, v \rangle + \langle v, u \rangle)(\|u\|^2 + \|v\|^2 + \langle u, -v \rangle + \langle -v, u \rangle)} \\ &= \sqrt{(\|u\|^2 + \|v\|^2 + (\langle u, v \rangle + \langle v, u \rangle))(\|u\|^2 + \|v\|^2 - (\langle u, v \rangle + \langle v, u \rangle))}\end{aligned}$$

let $a = \|u\|^2 + \|v\|^2$ and $b = \langle u, v \rangle + \langle v, u \rangle$, then

$$\begin{aligned}&\|u + v\|\|u - v\| \\ &= \sqrt{(a + b)(a - b)} \\ &= \sqrt{a^2 - b^2}\end{aligned}$$

while the right hand side is: $\|u\|^2 + \|v\|^2 = a$, then square both sides we have:

$$a^2 - b^2 \leq a^2$$

which is obviously true, cause $b^2 \geq 0$ (it is not the case that b is a complex number, as $b = \|u + v\|^2 - \|u\|^2 - \|v\|^2$). $\square$

Exercise 6.23. Let $e_0, \dots, e_{m-1}$ a list of vectors in V , such that

$$\|a_0e_0 + \dots + a_{m-1}e_{m-1}\|^2 = |a_0|^2 + \dots + |a_{m-1}|^2$$

for any $a_0, \dots, a_{m-1} \in V$. Show that $e_0, \dots, e_{m-1}$ is an orthonormal basis.

Proof. For any $i = 0, \dots, m-1$, we have $\|e_i\|^2 = |1|^2$ if we pick $a_i = 1$ and others = 0.

For any $i, j = 0, \dots, m-1$ and $i \neq j$, we have $\|a_ie_i + a_j e_j\|^2 = |a_i|^2 + |a_j|^2$. Then

$$\begin{aligned} & \|a_ie_i + a_j e_j\|^2 \\ &= \langle a_ie_i + a_j e_j, a_ie_i + a_j e_j \rangle \\ &= \langle a_ie_i, a_ie_i \rangle + \langle a_j e_j, a_j e_j \rangle + \langle a_ie_i, a_j e_j \rangle + \langle a_j e_j, a_ie_i \rangle \\ &= a_i \bar{a}_i \langle e_i, e_i \rangle + a_j \bar{a}_j \langle e_j, e_j \rangle + a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \langle e_j, e_i \rangle \\ &= a_i \bar{a}_i + a_j \bar{a}_j + a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \langle e_j, e_i \rangle \\ &= |a_i|^2 + |a_j|^2 + a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \overline{\langle e_i, e_j \rangle} \\ &= |a_i|^2 + |a_j|^2 \end{aligned}$$

that means $a_i \bar{a}_j \langle e_i, e_j \rangle + a_j \bar{a}_i \overline{\langle e_i, e_j \rangle} = 0$. Pick $a_i = a_j = 1$ we have $2 \operatorname{Re} \langle e_i, e_j \rangle = 0$ and pick $a_i = i, a_j = 1$ we have $-2 \operatorname{Im} \langle e_i, e_j \rangle = 0$. $\square$

Exercise 6.24.

- Let $\theta \in R$, show that:

$$(\cos \theta, \sin \theta), (-\sin \theta, \cos \theta) \quad \text{and} \quad (\cos \theta, \sin \theta), (\sin \theta, -\cos \theta)$$

are orthonormal basis of R^2 .

- Show that any orthogonal basis of R^2 is in one of such form.

Proof.

- They are normal cause $\cos^2 \theta + \sin^2 \theta = 1$, and orthogonal cause $-\cos \theta \sin \theta + \sin \theta \cos \theta = \cos \theta \sin \theta - \sin \theta \cos \theta = 0$.
- Any vector with length 1 in R^2 must in form of $(\cos \theta, \sin \theta)$. For any orthonormal basis, suppose it is $(\cos \theta, \sin \theta), (\cos \alpha, \sin \alpha)$, we may

write α as $\beta - \frac{\pi}{2}$, then the inner product of them is $\cos \theta \cos(\beta - \frac{\pi}{2}) + \sin \theta \sin(\beta - \frac{\pi}{2})$. Recall that $\sin \theta = \cos(\theta - \frac{\pi}{2})$ for any θ , then

$$\begin{aligned}
& \cos \theta \cos(\beta - \frac{\pi}{2}) + \sin \theta \sin(\beta - \frac{\pi}{2}) \\
&= \cos \theta \sin \beta + \sin \theta \cos(\beta - \pi) \\
&= \cos \theta \sin \beta + \sin \theta (-\cos \beta) \\
&= \cos \theta \sin \beta - \sin \theta \cos \beta \\
&= \sin(\theta - \beta) \\
&= 0
\end{aligned}$$

That means $\theta + \beta = k\pi$, thus $\theta + \alpha = k\pi - \frac{\pi}{2}$. If k is odd, then $\cos \alpha = \cos(k\pi - \frac{\pi}{2} - \theta) = \cos(\pi - \frac{\pi}{2} - \theta) = \sin(\pi - \theta) = -\sin(-\theta) = \sin \theta$ and $\sin \alpha = \sin(k\pi - \frac{\pi}{2} - \theta) = \sin(\pi - \frac{\pi}{2} - \theta) = \sin(\frac{\pi}{2} - \theta) = \sin(-(\theta + \frac{\pi}{2})) = -\sin(\theta + \frac{\pi}{2}) = -\cos \theta$. Similar to the case that k is even.

□